

Human review packet: FalahatgarEtAl2017MaxingRanking

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: FalahatgarEtAl2017MaxingRanking

Paper id: FalahatgarEtAl2017MaxingRanking

Source version: NeurIPS 2017 published main paper and official supplement

Source record: audit/paper_statement_map.json

Rows: 23

Generated: 2026-09-07

Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper FalahatgarEtAl2017MaxingRanking --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

- [Governing Lean declarations \(163; supporting context\)](#)
- [Paper-specific semantic prerequisites \(26; not paper claims\)](#)
 - [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
 - [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
 - [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
 - [FalahatgarEtAl2017MaxingRanking.bordaScore](#)
 - [FalahatgarEtAl2017MaxingRanking.EpsilonBordaMaximum](#)
 - [FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking](#)
 - [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
 - [FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking](#)
 - [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
 - [FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule](#)
 - [FalahatgarEtAl2017MaxingRanking.PreferenceRanking](#)
 - [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
 - [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
 - [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
 - [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw](#)
 - [FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateOutputLaw](#)
 - [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeSourceLogAlgorithmOutputLaw](#)
 - [FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate](#)
 - [FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget](#)
 - [FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget](#)
 - [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw](#)
 - [FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw](#)
 - [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff](#)
 - [FalahatgarEtAl2017MaxingRanking.stoppedPruneRounds](#)
 - [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairEmpiricalGap](#)

- FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput
- Source claims (23)
 - lemma1_compareSpec
 - theorem2_seqEliminateSpec
 - lemma3_pickAnchorSpec
 - remark4_pickAnchorLinearCutoffSpec
 - lemma5_pruneSpec
 - theorem6_optMaximizeSpec
 - theorem7_rankingLowerBoundSpec
 - theorem8_bordaMaxingSpec
 - theorem9_bordaRankingSpec
 - lemma10_compareCapSpec
 - lemma11_compareLowerSpec
 - lemma12_compareUpperSpec
 - lemma13_pruneRetentionSpec
 - lemma14_pruneOneRoundSpec
 - lemma15_pruneMultiRoundSpec
 - lemma16_finalCheckAnchorSpec
 - lemma17_anchorPruneSpec
 - lemma18_optMaximizeTailSpec
 - lemma19_estimateProbabilitySpec
 - lemma20_strongTransitivityRankingSpec
 - lemma21_estimateBordaScoreSpec
 - theorem22_bordaRankingSpec
 - sstMaximumAndRankingExistenceSpec

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.PMFKernel

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (Input : Type u_1) (Output : Type u_2) => Input → PMF Output
```

AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_2 u_1)

value:
fun (State : Type u_1) (Outcome : Type u_2) => ℕ → AppliedModelingLib.PMFKernel State Outcome
```

Direct retained dependencies

- [AppliedModelingLib.PMFKernel](#)

AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (State : Type u_1) (Outcome : Type u_2) => ℕ → State → Outcome → State
```

AppliedModelingLib.PreferenceRL.ApproximatePreferenceWinner

Declaration location: reusable library

Lean declaration body

```
type:
{Arm : Type u_1} → (Arm → Arm → ℝ) → Arm → ℝ → Prop

value:
fun {Arm : Type u_1} (preferenceGap : Arm → Arm → ℝ) (selected : Arm) (error : ℝ) =>
  ∀ (competitor : Arm), -error ≤ preferenceGap selected competitor
```

AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw

Declaration location: reusable library

Lean declaration body

```
type:
{State : Type u_1} →
{Outcome : Type u_2} →
[Fintype State] →
[DecidableEq State] →
[Fintype Outcome] →
[DecidableEq Outcome] →
PMF State →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel State Outcome →
AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate State Outcome →
```

```
(bad : ℕ → State → Outcome → Prop) →
  [(queryIndex : ℕ) →
    (state : State) → (outcome : Outcome) → Decidable (bad queryIndex state outcome)] →
    ℕ → PMF (State × Bool)
```

```
value:
fun {State : Type u_1} {Outcome : Type u_2} [Fintype State] [DecidableEq State] [Fintype Outcome] [DecidableEq Outcome]
  (initialStateLaw : PMF State) (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel State Outcome)
  (advance : AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate State Outcome) (bad : ℕ → State → Outcome → Prop)
  [(queryIndex : ℕ) → (state : State) → (outcome : Outcome) → Decidable (bad queryIndex state outcome)] (a : ℕ) =>
  Nat.brecOn a (AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw_f initialStateLaw outcomeLaw advance bad)
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate](#)

AppliedModelingLib.Probability.binaryRatingScore

Declaration location: reusable library

Lean declaration body

```
type:
Bool → ℝ
```

```
value:
fun (b : Bool) => if b = true then 1 else 0
```

AppliedModelingLib.Probability.finiteIIDSampleLaw

Declaration location: reusable library

Lean declaration body

```
type:
{X : Type u_1} →
  [inst : MeasurableSpace X] → MeasureTheory.Measure X → (count : ℕ) → MeasureTheory.Measure (Fin count → X)
```

```
value:
fun {X : Type u_1} [MeasurableSpace X] (law : MeasureTheory.Measure X) (count : ℕ) =>
  MeasureTheory.Measure.pi fun (x : Fin count) => law
```

AppliedModelingLib.Probability.realBernoulliPMF

Declaration location: reusable library

Lean declaration body

```
type:
(p : ℝ) → 0 ≤ p → p ≤ 1 → PMF Bool
```

```
value:
fun (p : ℝ) (hp0 : 0 ≤ p) (hp1 : p ≤ 1) =>
  PMF.bernoulli (p, hp0) (AppliedModelingLib.Probability.realBernoulliPMF_proof_1 p hp0 hp1)
```

AppliedModelingLib.finCons

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → {k : ℕ} → α → (Fin k → α) → Fin (k + 1) → α
```

```
value:
fun {α : Type u_1} {k : ℕ} (head : α) (tail : Fin k → α) (a : Fin (k + 1)) =>
  match a with
  | (0, _h) => head
  | (n.succ, h) => tail (n, AppliedModelingLib.finCons._proof_1 n h)
```

AppliedModelingLib.finiteAvailableWeight

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → [DecidableEq α] → (α → ℝ) → Finset α → ℝ

value:
fun {α : Type u_1} [Fintype α] [DecidableEq α] (baseWeight : α → ℝ) (forbidden : Finset α) =>
  ∑ a : α, if a ∈ forbidden then 0 else baseWeight a
```

AppliedModelingLib.finiteFreshList

Declaration location: reusable library

Lean declaration body

```
type:
(α : Type u_1) → ℕ → Finset α → Type (max 0 u_1)

value:
fun (α : Type u_1) (k : ℕ) (forbidden : Finset α) =>
  { list : Fin k → α // Function.Injective list ∧ ∀ (slot : Fin k), list slot ∉ forbidden }
```

AppliedModelingLib.finiteFreshListCons

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} →
[inst : DecidableEq α] →
{k : ℕ} →
{forbidden : Finset α} →
(head : { a : α // a ∉ forbidden }) →
  AppliedModelingLib.finiteFreshList α k (insert (↑ head) forbidden) →
  AppliedModelingLib.finiteFreshList α (k + 1) forbidden

value:
fun {α : Type u_1} [DecidableEq α] {k : ℕ} {forbidden : Finset α} (head : { a : α // a ∉ forbidden })
  (tail : AppliedModelingLib.finiteFreshList α k (insert (↑ head) forbidden)) =>
  (AppliedModelingLib.finCons ↑ head ↑ tail, AppliedModelingLib.finiteFreshListCons._proof_1 head tail)
```

Direct retained dependencies

- [AppliedModelingLib.finCons](#)
- [AppliedModelingLib.finiteFreshList](#)

AppliedModelingLib.finiteFreshListNil

Declaration location: reusable library

Lean declaration body

```
type:
(α : Type u_1) → (forbidden : Finset α) → AppliedModelingLib.finiteFreshList α 0 forbidden

value:
fun (α : Type u_1) (forbidden : Finset α) =>
  (fun (i : Fin 0) => i.elim0, AppliedModelingLib.finiteFreshListNil._proof_1 α forbidden)
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)

AppliedModelingLib.finiteWeightedPMF

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [inst : Fintype α] → (weight : α → ℝ) → (∀ (a : α), 0 ≤ weight a) → 0 < ∑ a : α, weight a → PMF α

value:
fun {α : Type u_1} [Fintype α] (weight : α → ℝ) (hweight_nonneg : ∀ (a : α), 0 ≤ weight a)
  (htotal_pos : 0 < ∑ a : α, weight a) =>
  PMF.ofFintype (fun (a : α) => ENNReal.ofReal (weight a / ∑ b : α, weight b))
  (AppliedModelingLib.finiteWeightedPMF._proof_1 weight hweight_nonneg htotal_pos)
```

AppliedModelingLib.finiteWeightedPMFAvailable

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} →
[inst : Fintype α] →
[inst_1 : DecidableEq α] →
(baseWeight : α → ℝ) →
(forbidden : Finset α) →
(∀ (a : α), 0 ≤ baseWeight a) →
0 < AppliedModelingLib.finiteAvailableWeight baseWeight forbidden → PMF { a : α // a ∉ forbidden }

value:
fun {α : Type u_1} [Fintype α] [DecidableEq α] (baseWeight : α → ℝ) (forbidden : Finset α)
(hbase_nonneg : ∀ (a : α), 0 ≤ baseWeight a)
(havailable_pos : 0 < AppliedModelingLib.finiteAvailableWeight baseWeight forbidden) =>
AppliedModelingLib.finiteWeightedPMF (fun (a : { a : α // a ∉ forbidden }) => baseWeight ↑ a)
(AppliedModelingLib.finiteWeightedPMFAvailable._proof_1 baseWeight forbidden hbase_nonneg)
(AppliedModelingLib.finiteWeightedPMFAvailable._proof_2 baseWeight forbidden havailable_pos)
```

Direct retained dependencies

- [AppliedModelingLib.finiteAvailableWeight](#)
- [AppliedModelingLib.finiteWeightedPMF](#)

AppliedModelingLib.finiteWithoutReplacementPMF

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} →
[inst : Fintype α] →
[inst_1 : DecidableEq α] →
(baseWeight : α → ℝ) →
(∀ (a : α), 0 ≤ baseWeight a) →
(∀ (forbidden : Finset α),
forbidden.card < Fintype.card α → 0 < AppliedModelingLib.finiteAvailableWeight baseWeight forbidden) →
(k : ℕ) →
(forbidden : Finset α) →
forbidden.card + k ≤ Fintype.card α → PMF (AppliedModelingLib.finiteFreshList α k forbidden)

value:
fun {α : Type u_1} [Fintype α] [DecidableEq α] (baseWeight : α → ℝ) (hbase_nonneg : ∀ (a : α), 0 ≤ baseWeight a)
(havailable :
∀ (forbidden : Finset α),
forbidden.card < Fintype.card α → 0 < AppliedModelingLib.finiteAvailableWeight baseWeight forbidden)
(k : ℕ) (forbidden : Finset α) (a : forbidden.card + k ≤ Fintype.card α) =>
AppliedModelingLib.finiteWithoutReplacementPMF_unary baseWeight hbase_nonneg havailable (k, {forbidden, a})
```

Direct retained dependencies

- [AppliedModelingLib.finiteAvailableWeight](#)
- [AppliedModelingLib.finiteFreshList](#)
- [AppliedModelingLib.finiteFreshListCons](#)
- [AppliedModelingLib.finiteFreshListNil](#)
- [AppliedModelingLib.finiteWeightedPMFAvailable](#)

AppliedModelingLib.pmfExp

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (α → ℝ) → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (f : α → ℝ) => ∑ a : α, (μ a).toReal * f a
```

AppliedModelingLib.pmfProb

Declaration location: reusable library

Lean declaration body

```

type:
{α : Type u_1} → [Fintype α] → PMF α → (p : α → Prop) → [DecidablePred p] → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (p : α → Prop) [DecidablePred p] =>
  AppliedModelingLib.pmfExp μ fun (a : α) => if p a then 1 else 0

```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.PreferenceRL.pmfProbClassical

Declaration location: reusable library

Lean declaration body

```

type:
{Outcome : Type u_1} → [Fintype Outcome] → PMF Outcome → (Outcome → Prop) → ℝ

value:
fun {Outcome : Type u_1} [Fintype Outcome] (law : PMF Outcome) (event : Outcome → Prop) =>
  AppliedModelingLib.pmfProb law event

```

Direct retained dependencies

- [AppliedModelingLib.pmfProb](#)

AppliedModelingLib.uniformPMF

Declaration location: reusable library

Lean declaration body

```

type:
(α : Type u_1) → [Fintype α] → [Nonempty α] → PMF α

value:
fun (α : Type u_1) [Fintype α] [Nonempty α] =>
  PMF.ofFintype (fun (x : α) => (↑ (Fintype.card α))-1) (AppliedModelingLib.uniformPMF_proof_1 α)

```

FalahatgarEtAl2017MaxingRanking.CompareDecision

Declaration location: paper-local

Lean declaration body

```

type:
Type

constructor FalahatgarEtAl2017MaxingRanking.CompareDecision.lower:
FalahatgarEtAl2017MaxingRanking.CompareDecision

constructor FalahatgarEtAl2017MaxingRanking.CompareDecision.upper:
FalahatgarEtAl2017MaxingRanking.CompareDecision

```

FalahatgarEtAl2017MaxingRanking.EstimatedScoresSorted

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → [inst : Fintype Arm] → (Arm → ℝ) → (Fin (Fintype.card Arm) → Arm) → Prop

value:
fun {Arm : Type u_1} [Fintype Arm] (estimate : Arm → ℝ) (ranking : Fin (Fintype.card Arm) → Arm) =>
  ∀ (first second : Fin (Fintype.card Arm)), ↑first ≤ ↑second → estimate (ranking second) ≤ estimate (ranking first)

```

FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex

Declaration location: paper-local

Lean declaration body

```

type:
(Arm : Type u_1) → [DecidableEq Arm] → Type (max 0 u_1)

value:
fun (Arm : Type u_1) [DecidableEq Arm] => { pair : Sym2 Arm // ¬pair.IsDiag }

```

FalahatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate

Declaration location: paper-local

Lean declaration body

```

type:
{Ω : Type u_1} → (ℕ → Ω → ℝ) → ℕ → Ω → ℝ

value:
fun {Ω : Type u_1} (observation : ℕ → Ω → ℝ) (count : ℕ) (outcome : Ω) =>
  (Σ index ∈ Finset.range count, observation index outcome) / ↑count - 1 / 2

```

FalahatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint

Declaration location: paper-local

Lean declaration body

```

type:
ℝ → ℝ → ℝ

value:
fun (lower upper : ℝ) => (lower + upper) / 2

```

FalahatgarEtAl2017MaxingRanking.adaptiveConfidenceRadius

Declaration location: paper-local

Lean declaration body

```

type:
ℕ → ℝ → ℝ

value:
fun (count : ℕ) (delta : ℝ) => √(Real.log (4 * ↑count ^ 2 / delta) / (2 * ↑count))

```

FalahatgarEtAl2017MaxingRanking.adaptiveCompareStep

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → {Ω : Type u_2} → (Arm → Arm → ℕ → Ω → ℝ) → ℕ → ℝ → ℝ → Ω → Arm → Arm → Arm

value:
fun {Arm : Type u_1} {Ω : Type u_2} (observation : Arm → Arm → ℕ → Ω → ℝ) (count : ℕ) (epsilon delta : ℝ) (outcome : Ω)
  (incumbent challenger : Arm) =>
  match
    | FalahatgarEtAl2017MaxingRanking.adaptiveCompare (observation challenger incumbent) count 0 epsilon delta
      outcome with
    | FalahatgarEtAl2017MaxingRanking.CompareDecision.lower => incumbent
    | FalahatgarEtAl2017MaxingRanking.CompareDecision.upper => challenger

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)

FalahatgarEtAl2017MaxingRanking.AdaptiveCompareStepCallValid

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → {Ω : Type u_2} → (Arm → Arm → ℕ → Ω → ℝ) → ℕ → (Arm → Arm → ℝ) → ℝ → ℝ → Ω → Arm → Arm → Prop

value:
fun {Arm : Type u_1} {Ω : Type u_2} (observation : Arm → Arm → ℕ → Ω → ℝ) (count : ℕ) (preferenceGap : Arm → Arm → ℝ)
  (epsilon delta : ℝ) (outcome : Ω) (incumbent challenger : Arm) =>
  0 ≤
    preferenceGap
      (FalihatgarEtAl2017MaxingRanking.adaptiveCompareStep observation count epsilon delta outcome incumbent
        challenger)
    incumbent ∧
  -epsilon ≤
    preferenceGap
      (FalihatgarEtAl2017MaxingRanking.adaptiveCompareStep observation count epsilon delta outcome incumbent
        challenger)
    challenger
```

Direct retained dependencies

- [FalihatgarEtAl2017MaxingRanking.adaptiveCompareStep](#)

FalihatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime

Declaration location: paper-local

Lean declaration body

```
type:
{Ω : Type u_1} → (ℕ → Ω → ℝ) → ℕ → ℝ → ℝ → ℝ → Ω → ℕ

value:
fun {Ω : Type u_1} (observation : ℕ → Ω → ℝ) (count : ℕ) (lower upper delta : ℝ) (outcome : Ω) =>
  if hstop :
    ∃ time ∈ Finset.lcc 1 count,
      FalihatgarEtAl2017MaxingRanking.adaptiveConfidenceRadius time delta <
        |FalihatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate observation time outcome -
          FalihatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint lower upper| then
    Nat.find hstop
  else count
```

Direct retained dependencies

- [FalihatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate](#)
- [FalihatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint](#)
- [FalihatgarEtAl2017MaxingRanking.adaptiveConfidenceRadius](#)

FalihatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ → ℝ

value:
fun (delta : ℝ) (round : ℕ) => delta / 2 ^ (round + 2)
```

FalihatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw

Declaration location: paper-local

Lean declaration body

```
type:
{State : Type u_1} →
{Outcome : Type u_2} →
[Fintype State] →
[DecidableEq State] →
[Fintype Outcome] →
[DecidableEq Outcome] →
PMF State →
  AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel State Outcome →
  AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate State Outcome → ℕ → PMF State

value:
fun {State : Type u_1} {Outcome : Type u_2} [Fintype State] [DecidableEq State] [Fintype Outcome] [DecidableEq Outcome]
  (initialStateLaw : PMF State) (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel State Outcome)
  (advance : AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate State Outcome) (a : ℕ) =>
  Nat.brecOn a (FalihatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw._f initialStateLaw outcomeLaw advance)
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.AdaptiveStateUpdate](#)

FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → ℕ → Type (max u_1 0)

value:
fun (Arm : Type u_1) (batchSize : ℕ) => Arm × Fin batchSize
```

FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
(batchSize : ℕ) → Arm → ℕ → (FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm batchSize → Bool) → ℝ

value:
fun {Arm : Type u_1} (batchSize : ℕ) (arm : Arm) (sample : ℕ)
  (labelTable : FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm batchSize → Bool) =>
  if hsample : sample < batchSize then
    AppliedModelingLib.Probability.binaryRatingScore (labelTable (arm, (sample, hsample)))
  else 0
```

Direct retained dependencies

- [AppliedModelingLib.Probability.binaryRatingScore](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate](#)

FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
(winProbability : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability → Arm → PMF Bool

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] (winProbability : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (arm : Arm) =>
  (AppliedModelingLib.uniformPMF Arm).bind fun (opponent : Arm) =>
    AppliedModelingLib.Probability.realBernoulliPMF (winProbability arm opponent)
  (FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw._proof_1 winProbability hprobability arm opponent)
  (FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw._proof_2 winProbability hprobability arm opponent)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.realBernoulliPMF](#)
- [AppliedModelingLib.uniformPMF](#)
- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)

FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
[Nonempty Arm] →
(winProbability : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability →
(batchSize : ℕ) → PMF (FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm batchSize → Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] [Nonempty Arm] (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (batchSize : ℕ) =>
let marginal :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm batchSize → MeasureTheory.Measure Bool :=
fun (coordinate : FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm batchSize) =>
(FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw winProbability hprobability coordinate.1).toMeasure;
(MeasureTheory.Measure.pi marginal).toPMF
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation

Declaration location: paper-local

Lean declaration body

```
type:
(batchSize : ℕ) → ℕ → (Fin batchSize → Bool) → ℝ

value:
fun (batchSize sampleIndex : ℕ) (outcome : Fin batchSize → Bool) =>
if hindex : sampleIndex < batchSize then
AppliedModelingLib.Probability.binaryRatingScore (outcome (sampleIndex, hindex))
else 0
```

Direct retained dependencies

- [AppliedModelingLib.Probability.binaryRatingScore](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome

Declaration location: paper-local

Lean declaration body

```
type:
(Arm : Type u_1) → [Fintype Arm] → ℕ → Type (max 0 u_1)

value:
fun (Arm : Type u_1) [Fintype Arm] (maxBatch : ℕ) =>
(batchSize : Fin (maxBatch + 1)) × (↑Finset.univ → Fin ↑batchSize → Bool)
```

FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedEmbed

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] →
(maxBatch batchSize : ℕ) →
batchSize ≤ maxBatch →
(↑Finset.univ → Fin batchSize → Bool) →
FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (maxBatch batchSize : ℕ) (hbatch : batchSize ≤ maxBatch)
(table : ↑Finset.univ → Fin batchSize → Bool) =>
((batchSize, Nat.lt_succ_of_le hbatch), table)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype

Declaration location: paper-local

Lean declaration body

```
type:
(Arm : Type u_1) →
  [inst : Fintype Arm] →
    (maxBatch : ℕ) → Fintype (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch)

value:
fun (Arm : Type u_1) [Fintype Arm] (maxBatch : ℕ) =>
  Fintype.ofFinite (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)

FalahatgarEtAl2017MaxingRanking.centeredComparisonProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → (Arm → Arm → ℝ) → Arm → Arm → ℝ

value:
fun {Arm : Type u_1} (preferenceGap : Arm → Arm → ℝ) (challenger incumbent : Arm) =>
  1 / 2 + preferenceGap challenger incumbent
```

FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
  (preferenceGap : Arm → Arm → ℝ) →
    FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
      Arm → Arm → (batchSize : ℕ) → PMF (Fin batchSize → Bool)

value:
fun {Arm : Type u_1} (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (challenger incumbent : Arm) (batchSize : ℕ) =>
  let probability : ℝ :=
    FalahatgarEtAl2017MaxingRanking.centeredComparisonProbability preferenceGap challenger incumbent;
  let marginal : PMF Bool :=
    AppliedModelingLib.Probability.realBernoulliPMF probability
      (FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw._proof_1 preferenceGap hprobability challenger
        incumbent)
      (FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw._proof_2 preferenceGap hprobability challenger
        incumbent);
  (AppliedModelingLib.Probability.finiteIIDSampleLaw marginal.toMeasure batchSize).toPMF
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finiteIIDSampleLaw](#)
- [AppliedModelingLib.Probability.realBernoulliPMF](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.centeredComparisonProbability](#)

FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[DecidableEq Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
(active : Finset Arm) → Arm → (batchSize : ℕ) → PMF (↑active → Fin batchSize → Bool)

value:
fun {Arm : Type u_1} [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (active : Finset Arm)
(anchor : Arm) (batchSize : ℕ) =>
let marginal : ↑active → MeasureTheory.Measure (Fin batchSize → Bool) := fun (arm : ↑active) =>
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability (↑arm) anchor
batchSize).toMeasure;
(MeasureTheory.Measure.pi marginal).toPMF
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw](#)

FalahatgarEtAl2017MaxingRanking.estimatedThresholdCandidateSet

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → (Arm → Arm → ℝ) → ℝ → Finset Arm → Finset Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (estimate : Arm → Arm → ℝ) (epsilon : ℝ) (active : Finset Arm) =>
{selected ∈ active | ∀ competitor ∈ active, -epsilon / 2 ≤ estimate selected competitor}
```

FalahatgarEtAl2017MaxingRanking.finalCheckTraceState

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → Finset Arm → Type

value:
fun {Arm : Type u_1} [DecidableEq Arm] (candidates : Finset Arm) => (Fin candidates.card → Option Bool) × Bool
```

FalahatgarEtAl2017MaxingRanking.finsetFinalCheckCandidate

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → Arm → Finset Arm → ℕ → Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (anchor : Arm) (candidates : Finset Arm) (queryIndex : ℕ) =>
candidates.toList.getD queryIndex anchor
```

FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateChallengers

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → Finset Arm → List Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (candidates : Finset Arm) => candidates.toList.tail
```

FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateInitial

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → Arm → Finset Arm → Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (emptyFallback : Arm) (candidates : Finset Arm) =>
  candidates.toList.headD emptyFallback
```

FalahatgarEtAl2017MaxingRanking.fixedSampleBudget

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ 

value:
fun (lower upper delta :  $\mathbb{R}$ ) => [2 / (upper - lower) ^ 2 * Real.log (2 / delta)]+
```

FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision

Declaration location: paper-local

Lean declaration body

```
type:
{ $\Omega$  : Type u_1} →
{Arm : Type u_2} → ( $\mathbb{N} \rightarrow \text{Arm} \rightarrow \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) →  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \text{Arm} \rightarrow \Omega \rightarrow \text{FalahatgarEtAl2017MaxingRanking.CompareDecision}$ 

value:
fun { $\Omega$  : Type u_1} {Arm : Type u_2} (observation :  $\mathbb{N} \rightarrow \text{Arm} \rightarrow \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) (lower upper delta :  $\mathbb{R}$ ) (round :  $\mathbb{N}$ ) (arm : Arm)
  (outcome :  $\Omega$ ) =>
  FalahatgarEtAl2017MaxingRanking.adaptiveCompare (observation round arm)
  (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round))
  lower upper (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) outcome
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.bordaSampleBudget

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ 

value:
fun (armCount :  $\mathbb{N}$ ) (epsilon delta :  $\mathbb{R}$ ) =>
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon (delta / ↑armCount)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ 

value:
fun (armCount :  $\mathbb{N}$ ) (epsilon delta :  $\mathbb{R}$ ) =>
  armCount * FalahatgarEtAl2017MaxingRanking.bordaSampleBudget armCount epsilon delta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.bordaSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareObservation

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
(epsilon eta : ℝ) →
ℕ → Arm → Arm → ℕ → (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool) → ℝ

value:
fun {Arm : Type u_1} (epsilon eta : ℝ) (a : ℕ) (a_1 a_2 : Arm) (a_3 : ℕ)
(a_4 : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool) =>
FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) a_3 a_4
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareOutcomeLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
Arm →
List Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
(epsilon eta : ℝ) →
ℕ → Arm → PMF (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool)

value:
fun {Arm : Type u_1} (initial : Arm) (challengers : List Arm) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon eta : ℝ)
(a : ℕ) (a_1 : Arm) =>
FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability (challengers.getD a initial)
a_1 (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckAdvance

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
Finset Arm →
(lower upper eta : ℝ) →
ℕ → Bool → (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) → Bool

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (candidates : Finset Arm) (lower upper eta : ℝ) (a : ℕ)
(a_1 : Bool) (a_2 : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) =>
if hqueryIndex : a < candidates.card then
a_1 &&
decide
(FalahatgarEtAl2017MaxingRanking.adaptiveCompare
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta))
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) lower upper eta a_2 ≠
FalahatgarEtAl2017MaxingRanking.CompareDecision.upper)
else a_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutcomeLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
Arm →
Finset Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
(lower upper eta : ℝ) →
ℕ → Bool → PMF (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchor : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (lower upper eta : ℝ)
  (a : ℕ) (a_1 : Bool) =>
FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.finsetFinalCheckCandidate anchor candidates a) anchor
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.finsetFinalCheckCandidate](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckNoUpperLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
Arm →
Finset Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → ℝ → PMF Bool

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchor : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (lower upper eta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw (PMF.pure true)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutcomeLaw anchor candidates preferenceGap hprobability
 lower upper eta)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckAdvance candidates lower upper eta) candidates.card
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
Arm →
Arm →
Finset Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → ℝ → PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchor fallback : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (lower upper eta : ℝ) =>
PMF.map (fun (noUpper : Bool) => if noUpper = true then anchor else fallback)
  (FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckNoUpperLaw anchor candidates preferenceGap hprobability
    lower upper eta)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckNoUpperLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceAdvance

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[inst : DecidableEq Arm] →
(candidates : Finset Arm) →
(lower upper eta : ℝ) →
ℕ →
FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates →
(Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) →
FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (candidates : Finset Arm) (lower upper eta : ℝ) (a : ℕ)
  (a_1 : FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates)
  (a_2 : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) =>
if hqueryIndex : a < candidates.card then
  if hnoUpper : a_1.2 = true then
    have isUpper : Bool :=
    decide
    (FalahatgarEtAl2017MaxingRanking.adaptiveCompare
      (FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta))
      (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) lower upper eta a_2 =
      FalahatgarEtAl2017MaxingRanking.CompareDecision.upper);
    (Function.update a_1.1 (a, hqueryIndex) (some isUpper), !isUpper)
  else a_1
else a_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{count : ℕ} →
  AppliedModelingLib.finiteFreshList Arm count 0 →
    (epsilon eta : ℝ) →
      ℕ → Arm → Arm → ℕ → (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool) → ℝ

value:
fun {Arm : Type u_1} {count : ℕ} ( _sample : AppliedModelingLib.finiteFreshList Arm count 0) (epsilon eta : ℝ) (a : ℕ)
(a_1 a_2 : Arm) (a_3 : ℕ) (a_4 : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool) =>
  FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareObservation epsilon eta a a_1 a_2 a_3 a_4
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareObservation](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] →
  (preferenceGap : Arm → Arm → ℝ) →
    FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
      (roundCount : ℕ) →
        Arm →
          (lower upper delta : ℝ) →
            (maxBatch : ℕ) →
              (∀ round < roundCount,
                FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
                  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
                    maxBatch) →
                ℕ → Finset Arm → PMF (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (roundCount : ℕ)
(anchor : Arm) (lower upper delta : ℝ) (maxBatch : ℕ)
(hbudget :
  ∀ round < roundCount,
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
        maxBatch)
(a : ℕ) (a_1 : Finset Arm) =>
  if hround : a < roundCount then
    PMF.map
      (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedEmbed maxBatch
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta a))
        (hbudget a hround))
      (FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw preferenceGap hprobability Finset.univ anchor
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta a)))
  else
    PMF.pure
      ((0, Nat.succ_pos maxBatch), fun (x : 1 Finset.univ) (sample : Fin ↑(0, Nat.succ_pos maxBatch)) => sample.elim0)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedEmbed](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[DecidableEq Arm] →
(active : Finset Arm) →
(lower upper eta : ℝ) →
(↑active → Fin (FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) →
Arm → FalihatgarEtAl2017MaxingRanking.CompareDecision

value:
fun {Arm : Type u_1} [DecidableEq Arm] (active : Finset Arm) (lower upper eta : ℝ)
(batchTable : ↑active → Fin (FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool)
(arm : Arm) =>
if hmem : arm ∈ active then
FalihatgarEtAl2017MaxingRanking.adaptiveCompare
(fun (sampleIndex : ℕ) (batch : Fin (FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) =>
FalihatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
(FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) sampleIndex batch)
(FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) lower upper eta (batchTable (arm, hmem))
else FalihatgarEtAl2017MaxingRanking.CompareDecision.lower
```

Direct retained dependencies

- [FalihatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalihatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalihatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalihatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalihatgarEtAl2017MaxingRanking.canonicalFullPruneRoundDecision

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] →
(lower upper eta : ℝ) →
(↑Finset.univ → Fin (FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool) →
Arm → FalihatgarEtAl2017MaxingRanking.CompareDecision

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (lower upper eta : ℝ)
(batchTable : ↑Finset.univ → Fin (FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool)
(arm : Arm) =>
FalihatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision Finset.univ lower upper eta batchTable arm
```

Direct retained dependencies

- [FalihatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalihatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision](#)
- [FalihatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalihatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] →
(maxBatch : ℕ) →
(lower upper eta : ℝ) →
FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta ≤ maxBatch →
FalihatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch →
Arm → FalihatgarEtAl2017MaxingRanking.CompareDecision

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (maxBatch : ℕ) (lower upper eta : ℝ)
(hbudget : FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta ≤ maxBatch)
(outcome : FalihatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch) (arm : Arm) =>
if htag : ↑outcome.fst = FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta then
have htagFin :
outcome.fst =
(FalihatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta,
```

```

    FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision._proof_1 maxBatch lower upper eta
    hbudget) :=
FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision._proof_2 maxBatch lower upper eta hbudget
outcome htag;
have batchTable : 1 Finset.univ → Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta) → Bool :=
cast
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision._proof_3 maxBatch lower upper eta hbudget
outcome htagFin)
outcome.snd;
FalahatgarEtAl2017MaxingRanking.canonicalFullPruneRoundDecision lower upper eta batchTable arm
else FalahatgarEtAl2017MaxingRanking.CompareDecision.lower

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFullPruneRoundDecision](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] →
(roundCount : ℕ) →
(lower upper delta : ℝ) →
(maxBatch : ℕ) →
(∀ round < roundCount,
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
  maxBatch) →
ℕ →
Finset Arm →
  FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch →
  Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (roundCount : ℕ) (lower upper delta : ℝ) (maxBatch : ℕ)
  (hbudget :
    ∀ round < roundCount,
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
      maxBatch)
  (round : ℕ) (active : Finset Arm)
  (outcome : FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch) (arm : Arm) =>
if hround : round < roundCount then
  FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision maxBatch lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) (hbudget round hround) outcome arm
else FalahatgarEtAl2017MaxingRanking.CompareDecision.lower

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.finalCheckTraceComparisonCount

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → [DecidableEq Arm] → (candidates : Finset Arm) → ℝ → ℝ → ℝ → (Fin candidates.card → Option Bool) → ℕ

value:
fun {Arm : Type u_1} [DecidableEq Arm] (candidates : Finset Arm) (lower upper eta : ℝ)
  (trace : Fin candidates.card → Option Bool) =>
{queryIndex : Fin candidates.card | (trace queryIndex).isSome = true}.card *
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.freshSeqEliminateStateLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[DecidableEq Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
Arm →
List Arm →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome →
(ℕ → Arm → Arm → Outcome → Arm) →
(bad : ℕ → Arm → Outcome → Prop) →
[(queryIndex : ℕ) →
  (incumbent : Arm) → (outcome : Outcome) → Decidable (bad queryIndex incumbent outcome)] →
PMF (Arm × Bool)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [DecidableEq Arm] [Fintype Outcome] [DecidableEq Outcome]
  (initial : Arm) (challengers : List Arm)
  (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome)
  (step : ℕ → Arm → Arm → Outcome → Arm) (bad : ℕ → Arm → Outcome → Prop)
  [(queryIndex : ℕ) → (incumbent : Arm) → (outcome : Outcome) → Decidable (bad queryIndex incumbent outcome)] =>
AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw (PMF.pure initial) outcomeLaw
  (fun (queryIndex : ℕ) (incumbent : Arm) (outcome : Outcome) =>
    step queryIndex incumbent (challengers.getD queryIndex initial) outcome)
  bad challengers.length
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw](#)

FalahatgarEtAl2017MaxingRanking.freshAdaptiveCompareSeqEliminateStateLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
Arm →
List Arm →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome →
(ℕ → Arm → Arm → ℕ → Outcome → ℝ) → (Arm → Arm → ℝ) → ℝ → ℝ → PMF (Arm × Bool)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [Fintype Outcome] [DecidableEq Outcome] (initial : Arm)
  (challengers : List Arm) (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome)
  (observation : ℕ → Arm → Arm → ℕ → Outcome → ℝ) (preferenceGap : Arm → Arm → ℝ) (epsilon eta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.freshSeqEliminateStateLaw initial challengers outcomeLaw
  (fun (queryIndex : ℕ) (incumbent challenger : Arm) (outcome : Outcome) =>
    FalahatgarEtAl2017MaxingRanking.adaptiveCompareStep (observation queryIndex)
      (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) epsilon eta outcome incumbent challenger)
  fun (queryIndex : ℕ) (incumbent : Arm) (outcome : Outcome) =>
    if x : queryIndex < challengers.length then
      ¬FalahatgarEtAl2017MaxingRanking.AdaptiveCompareStepCallValid (observation queryIndex)
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) preferenceGap epsilon eta outcome incumbent
        (challengers.getD queryIndex initial)
    else False
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [FalahatgarEtAl2017MaxingRanking.AdaptiveCompareStepCallValid](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompareStep](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshSeqEliminateStateLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
Arm →
Finset Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] (emptyFallback : Arm) (candidates : Finset Arm) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon eta : ℝ) =>
PMF.map Prod.fst
(FalahatgarEtAl2017MaxingRanking.freshAdaptiveCompareSeqEliminateStateLaw
(FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateInitial emptyFallback candidates)
(FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateChallengers candidates)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareOutcomeLaw
(FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateInitial emptyFallback candidates)
(FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateChallengers candidates) preferenceGap hprobability epsilon
eta)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareObservation epsilon eta) preferenceGap epsilon eta)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareObservation](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateChallengers](#)
- [FalahatgarEtAl2017MaxingRanking.finsetSeqEliminateInitial](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshAdaptiveCompareSeqEliminateStateLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw
(Classical.choice FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw._proof_1) Finset.univ
preferenceGap hprobability epsilon (delta / ↑(Fintype.card Arm))
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTailSourceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
Arm →
Finset Arm →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → PMF (Arm × Arm)

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (anchor : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (epsilon delta : ℝ) =>
(FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw anchor candidates preferenceGap
  hprobability epsilon (delta / 4 / ↑ candidates.card)).bind
fun (fallback : Arm) =>
PMF.map (fun (output : Arm) => (fallback, output))
(FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw anchor fallback candidates preferenceGap
  hprobability (2 * epsilon / 3) epsilon (delta / 4 / ↑ (Fintype.card Arm)))
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw](#)

FalahatgarEtAl2017MaxingRanking.initialFinalCheckTraceState

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : DecidableEq Arm] → (candidates : Finset Arm) → FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates

value:
fun {Arm : Type u_1} [DecidableEq Arm] (candidates : Finset Arm) => (fun (x : Fin candidates.card) => none, true)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[inst : DecidableEq Arm] →
Arm →
(candidates : Finset Arm) →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
ℝ → ℝ → ℝ → PMF (FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchor : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (lower upper eta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw
(PMF.pure (FalahatgarEtAl2017MaxingRanking.initialFinalCheckTraceState candidates))
(fun (queryIndex : ℕ) (state : FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates) =>
  FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutcomeLaw anchor candidates preferenceGap hprobability
  lower upper eta queryIndex state.2)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceAdvance candidates lower upper eta) candidates.card
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.initialFinalCheckTraceState](#)

FalahatgarEtAl2017MaxingRanking.meanEstimateSampleBudget

Declaration location: paper-local

Lean declaration body

type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$

value:
fun (epsilon delta : $\mathbb{R}$) => FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 (2 * epsilon) delta

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneLaw

Declaration location: paper-local

Lean declaration body

type:
{Arm : Type u_1} → [Fintype Arm] → [DecidableEq Arm] → PMF Arm → (Arm → PMF (Finset Arm)) → PMF (Arm × Finset Arm)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchorLaw : PMF Arm) (pruneLaw : Arm → PMF (Finset Arm)) =>
 anchorLaw.bind fun (anchor : Arm) => PMF.map (fun (candidates : Finset Arm) => (anchor, candidates)) (pruneLaw anchor)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackLaw

Declaration location: paper-local

Lean declaration body

type:
{Arm : Type u_1} →
 [Fintype Arm] →
 [DecidableEq Arm] →
 PMF Arm → (Arm → PMF (Finset Arm)) → (Arm → Finset Arm → PMF Arm) → PMF ((Arm × Finset Arm) × Arm)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchorLaw : PMF Arm) (pruneLaw : Arm → PMF (Finset Arm))
 (fallbackLaw : Arm → Finset Arm → PMF Arm) =>
 (FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneLaw anchorLaw pruneLaw).bind
 fun (anchorCandidates : Arm × Finset Arm) =>
 PMF.map (fun (fallback : Arm) => (anchorCandidates, fallback)) (fallbackLaw anchorCandidates.1 anchorCandidates.2)

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneLaw](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackFinalLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
PMF Arm →
(Arm → PMF (Finset Arm)) →
(Arm → Finset Arm → PMF Arm) → (Arm → Finset Arm → Arm → PMF Arm) → PMF (((Arm × Finset Arm) × Arm) × Arm)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (anchorLaw : PMF Arm) (pruneLaw : Arm → PMF (Finset Arm))
(fallbackLaw : Arm → Finset Arm → PMF Arm) (finalLaw : Arm → Finset Arm → Arm → PMF Arm) =>
(FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackLaw anchorLaw pruneLaw fallbackLaw).bind
fun (anchorCandidatesFallback : (Arm × Finset Arm) × Arm) =>
PMF.map (fun (output : Arm) => (anchorCandidatesFallback, output))
(finalLaw anchorCandidatesFallback.1.1 anchorCandidatesFallback.1.2 anchorCandidatesFallback.2)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackLaw](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → Finset Arm → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} (candidates : Finset Arm) (epsilon delta : ℝ) =>
(candidates.card - 1) * FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon (delta / ↑candidates.card)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeBaseComparisonCap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} [Fintype Arm] (epsilon delta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap Finset.univ epsilon delta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceJointResourceProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ) =>
AppliedModelingLib.PreferenceRL.pmfProbClassical
(FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw preferenceGap hprobability epsilon delta)
fun (output : Arm) =>
FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap epsilon output ^
↑(FalahatgarEtAl2017MaxingRanking.optMaximizeBaseComparisonCap epsilon delta) ≤
↑(Fintype.card Arm - 1) * (2 / epsilon ^ 2 * Real.log (2 / (delta / ↑(Fintype.card Arm)))) + 1
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeBaseComparisonCap](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeFinalCheckComparisonCap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → Finset Arm → ℝ → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} (candidates : Finset Arm) (lower upper eta : ℝ) =>
  candidates.card * FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper eta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailAlgorithm3SourceBudget

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] → Finset Arm → ℝ → ℝ → ℝ → ℝ → ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] (candidates : Finset Arm) (lower upper epsilon delta : ℝ) =>
  if candidates.Nonempty then
    ↑ candidates.card * (2 / (upper - lower) ^ 2 * (Real.log (8 / delta) + Real.log ↑ (Fintype.card Arm)) + 1) +
    ↑ (candidates.card - 1) * (2 / epsilon ^ 2 * (Real.log (8 / delta) + Real.log ↑ candidates.card) + 1)
  else 0
```

FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailComparisonCap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → Finset Arm → ℝ → ℝ → ℝ → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} (candidates : Finset Arm) (lower upper finalEta fallbackEpsilon fallbackDelta : ℝ) =>
  FalahatgarEtAl2017MaxingRanking.optMaximizeFinalCheckComparisonCap candidates lower upper finalEta +
  FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap candidates fallbackEpsilon fallbackDelta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeFinalCheckComparisonCap](#)

FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ → ℕ

value:
fun (armCount cutoff : ℕ) (delta : ℝ) => min armCount [↑ armCount / ↑ cutoff * Real.log (2 / delta)] +
```

FalahatgarEtAl2017MaxingRanking.optMaximizePickAnchorComparisonCap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] → ℕ → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} [Fintype Arm] (cutoff : ℕ) (anchorDelta lower : ℝ) =>
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff anchorDelta - 1) *
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 lower
(anchorDelta / 2 / ↑(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff anchorDelta))
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)

FalahatgarEtAl2017MaxingRanking.pickAnchorSampleInitial

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → {count : ℕ} → AppliedModelingLib.finiteFreshList Arm count ∅ → 0 < count → Arm

value:
fun {Arm : Type u_1} {count : ℕ} (sample : AppliedModelingLib.finiteFreshList Arm count ∅) (hcount : 0 < count) =>
↑sample (0, hcount)
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)

FalahatgarEtAl2017MaxingRanking.pickAnchorSampleList

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → {count : ℕ} → AppliedModelingLib.finiteFreshList Arm count ∅ → List Arm

value:
fun {Arm : Type u_1} {count : ℕ} (sample : AppliedModelingLib.finiteFreshList Arm count ∅) => List.ofFn ↑sample
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)

FalahatgarEtAl2017MaxingRanking.pickAnchorSampleChallengers

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → {count : ℕ} → AppliedModelingLib.finiteFreshList Arm count ∅ → List Arm

value:
fun {Arm : Type u_1} {count : ℕ} (sample : AppliedModelingLib.finiteFreshList Arm count ∅) =>
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleList sample).tail
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleList](#)

FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
{count : ℕ} →
PMF (AppliedModelingLib.finiteFreshList Arm count 0) →
(AppliedModelingLib.finiteFreshList Arm count 0 →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome) →
(AppliedModelingLib.finiteFreshList Arm count 0 → ℕ → Arm → Arm → ℕ → Outcome → ℝ) →
(Arm → Arm → ℝ) →
ℝ → ℝ → 0 < count → PMF (AppliedModelingLib.finiteFreshList Arm count 0 × Arm × Bool)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [Fintype Outcome] [DecidableEq Outcome] {count : ℕ}
(sampleLaw : PMF (AppliedModelingLib.finiteFreshList Arm count 0))
(outcomeLaw :
AppliedModelingLib.finiteFreshList Arm count 0 →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome)
(observation : AppliedModelingLib.finiteFreshList Arm count 0 → ℕ → Arm → Arm → ℕ → Outcome → ℝ)
(preferenceGap : Arm → Arm → ℝ) (epsilon eta : ℝ) (hcount : 0 < count) =>
sampleLaw.bind fun (sample : AppliedModelingLib.finiteFreshList Arm count 0) =>
PMF.map (fun (stateFailure : Arm × Bool) => (sample, stateFailure))
(FalahatgarEtAl2017MaxingRanking.freshAdaptiveCompareSeqEliminateStateLaw
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleInitial sample hcount)
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleChallengers sample) (outcomeLaw sample) (observation sample)
preferenceGap epsilon eta)
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.freshAdaptiveCompareSeqEliminateStateLaw](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleChallengers](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleInitial](#)

FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointGoodAnchorProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
{count : ℕ} →
PMF (AppliedModelingLib.finiteFreshList Arm count 0) →
(AppliedModelingLib.finiteFreshList Arm count 0 →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome) →
(AppliedModelingLib.finiteFreshList Arm count 0 → ℕ → Arm → Arm → ℕ → Outcome → ℝ) →
(Arm → Arm → ℝ) →
ℝ → ℝ → ℕ → 0 < count → ℝ

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [Fintype Outcome] [DecidableEq Outcome] {count : ℕ}
(sampleLaw : PMF (AppliedModelingLib.finiteFreshList Arm count 0))
(outcomeLaw :
AppliedModelingLib.finiteFreshList Arm count 0 →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome)
(observation : AppliedModelingLib.finiteFreshList Arm count 0 → ℕ → Arm → Arm → ℕ → Outcome → ℝ)
(preferenceGap : Arm → Arm → ℝ) (epsilon eta : ℝ) (cutoff : ℕ) (hcount : 0 < count) =>
AppliedModelingLib.PreferenceRL.pmfProbClassical
(FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw sampleLaw outcomeLaw observation preferenceGap epsilon eta
hcount)
fun (sampleStateFailure : AppliedModelingLib.finiteFreshList Arm count 0 × Arm × Bool) =>
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap epsilon cutoff sampleStateFailure.2.1
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
(cutoff : ℕ) →
(delta : ℝ) →
ℝ →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
0 < cutoff → 0 < delta → delta ≤ 1 → PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] (cutoff : ℕ) (delta epsilon : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hcutoff : 0 < cutoff) (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1) =>
let count : ℕ := FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta;
have hcount : 0 < count :=
FalahatgarEtAl2017MaxingRanking.pickAnchorSourceSampleCount_pos cutoff delta hcutoff hdelta hdeltaLeOne;
PMF.map
(fun (sampleStateFailure : AppliedModelingLib.finiteFreshList Arm count ∅ × Arm × Bool) => sampleStateFailure.2.1)
(FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw
(FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw count
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw._proof_2 cutoff delta))
(fun (sample : AppliedModelingLib.finiteFreshList Arm count ∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw sample hcount preferenceGap hprobability
epsilon (delta / 2 / ↑count))
(fun (sample : AppliedModelingLib.finiteFreshList Arm count ∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation sample epsilon (delta / 2 / ↑count))
preferenceGap epsilon (delta / 2 / ↑count) hcount)
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateJointLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
ℝ → ℝ → PMF (AppliedModelingLib.finiteFreshList Arm (Fintype.card Arm) ∅ × Arm × Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw
(FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw (Fintype.card Arm)
FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateJointLaw._proof_1)
(fun (sample : AppliedModelingLib.finiteFreshList Arm (Fintype.card Arm) ∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw sample Fintype.card_pos preferenceGap
hprobability epsilon (delta / ↑(Fintype.card Arm)))
(fun (sample : AppliedModelingLib.finiteFreshList Arm (Fintype.card Arm) ∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation sample epsilon (delta / ↑(Fintype.card Arm)))
preferenceGap epsilon (delta / ↑(Fintype.card Arm)) Fintype.card_pos
```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointLaw](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw](#)

FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateEpsilonMaximumProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(preferenceGap : Arm → Arm → ℝ) →
  FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (epsilon delta : ℝ) =>
  AppliedModelingLib.PreferenceRL.pmfProbClassical
  (FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateOutputLaw preferenceGap hprobability epsilon
  delta)
  (FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap epsilon)
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateOutputLaw](#)

FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
(cutoff : ℕ) →
(delta : ℝ) →
  ℝ →
  (AppliedModelingLib.finiteFreshList Arm
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta) ∅ →
  AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome) →
  (AppliedModelingLib.finiteFreshList Arm
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta) ∅ →
  ℕ → Arm → Arm → ℕ → Outcome → ℝ) →
  (Arm → Arm → ℝ) → 0 < cutoff → 0 < delta → delta ≤ 1 → ℝ

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [Nonempty Arm] [Fintype Outcome] [DecidableEq Outcome]
  (cutoff : ℕ) (delta epsilon : ℝ)
  (outcomeLaw :
  AppliedModelingLib.finiteFreshList Arm
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta) ∅ →
  AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel Arm Outcome)
  (observation :
  AppliedModelingLib.finiteFreshList Arm
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta) ∅ →
  ℕ → Arm → Arm → ℕ → Outcome → ℝ)
  (preferenceGap : Arm → Arm → ℝ) (hcutoff : 0 < cutoff) (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1) =>
```

```
FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointGoodAnchorProbability
(FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)
(FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability._proof_1 cutoff delta))
outcomeLaw observation preferenceGap epsilon
(delta / 2 / ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)) cutoff
(FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability._proof_3 cutoff delta hcutoff hdelta
hdeltaLeOne)
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.freshPickAnchorJointGoodAnchorProbability](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw](#)

FalahatgarEtAl2017MaxingRanking.pruneBadArms

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → (Arm → Arm → ℝ) → ℝ → Arm → Finset Arm → Finset Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ) (lower : ℝ) (anchor : Arm)
(active : Finset Arm) =>
{arm ∈ active | preferenceGap arm anchor ≤ lower}
```

FalahatgarEtAl2017MaxingRanking.pruneRound

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [DecidableEq Arm] → Finset Arm → (Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) → Finset Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (active : Finset Arm)
(decision : Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) =>
{arm ∈ active | decision arm = FalahatgarEtAl2017MaxingRanking.CompareDecision.upper}
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)

FalahatgarEtAl2017MaxingRanking.pruneRounds

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[DecidableEq Arm] → List (Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) → Finset Arm → Finset Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (a : List (Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision))
(a_1 : Finset Arm) =>
List.brecOn (motive := fun (x : List (Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision)) =>
Finset Arm → Finset Arm) a FalahatgarEtAl2017MaxingRanking.pruneRounds._f a_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.pruneRound](#)

FalahatgarEtAl2017MaxingRanking.indexedPruneRounds

Declaration location: paper-local

Lean declaration body

```
type:
{Ω : Type u_1} →
{Arm : Type u_2} →
[DecidableEq Arm] →
ℕ → (ℕ → Arm → Ω → FalahatgarEtAl2017MaxingRanking.CompareDecision) → Ω → Finset Arm → Finset Arm

value:
fun {Ω : Type u_1} {Arm : Type u_2} [DecidableEq Arm] (roundCount : ℕ)
  (decision : ℕ → Arm → Ω → FalahatgarEtAl2017MaxingRanking.CompareDecision) (outcome : Ω) (active : Finset Arm) =>
  FalahatgarEtAl2017MaxingRanking.pruneRounds
  (List.map (fun (index : ℕ) (arm : Arm) => decision index arm outcome) (List.range roundCount)) active
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.pruneRounds](#)

FalahatgarEtAl2017MaxingRanking.selectEstimatedThresholdCandidate

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Nonempty Arm] → [DecidableEq Arm] → (Arm → Arm → ℝ) → ℝ → Finset Arm → Arm

value:
fun {Arm : Type u_1} [Nonempty Arm] [DecidableEq Arm] (estimate : Arm → Arm → ℝ) (epsilon : ℝ) (active : Finset Arm) =>
  if hcandidate : (FalahatgarEtAl2017MaxingRanking.estimatedThresholdCandidateSet estimate epsilon active).Nonempty then
    Exists.choose hcandidate
  else Classical.choice FalahatgarEtAl2017MaxingRanking.selectEstimatedThresholdCandidate._proof_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.estimatedThresholdCandidateSet](#)

FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ

value:
fun (armCount : ℕ) (epsilon delta : ℝ) => ↑(armCount - 1) * (2 / epsilon ^ 2 * Real.log (2 * ↑armCount / delta) + 1)
```

FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ → ℝ

value:
fun (armCount : ℕ) (lower upper delta : ℝ) =>
  16 * ↑armCount * (2 / (upper - lower) ^ 2 + 1) * (1 + Real.log (1 / delta))
```

FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (armCount cutoff :  $\mathbb{N}$ ) (delta :  $\mathbb{R}$ ) => max delta (↑ cutoff / ↑ armCount)
```

FalahatgarEtAl2017MaxingRanking.sourceOptMaximizePickAnchorComparisonEnvelope

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (armCount :  $\mathbb{N}$ ) (lower anchorDelta :  $\mathbb{R}$ ) =>
  ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount armCount
    (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff armCount) anchorDelta -
    1) *
  (2 / lower ^ 2 *
    Real.log
      (4 *
        ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount armCount
          (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff armCount) anchorDelta) /
        anchorDelta) +
    1)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff](#)

FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (armCount :  $\mathbb{N}$ ) (lower upper epsilon delta :  $\mathbb{R}$ ) =>
  FalahatgarEtAl2017MaxingRanking.sourceOptMaximizePickAnchorComparisonEnvelope armCount lower (delta / 4) +
  FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound armCount lower upper (delta / 4) +
  2 * ↑ (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff armCount) *
  (2 / (upper - lower) ^ 2 * (Real.log (8 / delta) + Real.log ↑ armCount) + 1) +
  2 * ↑ (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff armCount) *
  (2 / epsilon ^ 2 * (Real.log (8 / delta) + Real.log ↑ armCount) + 1)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizePickAnchorComparisonEnvelope](#)

FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3TotalRateEnvelope

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (armCount :  $\mathbb{N}$ ) (epsilon delta :  $\mathbb{R}$ ) =>
  if delta ≤ 1 / ↑ armCount then
    FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope armCount epsilon delta
  else
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope armCount (epsilon / 3) (2 * epsilon / 3)
    epsilon delta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope](#)

FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{N}$ 

value:
fun (armCount :  $\mathbb{N}$ ) => Nat.log 2 armCount ^ 2 - 1
```

FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] →  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ 

value:
fun {Arm : Type u_1} [Fintype Arm] (lower upper delta :  $\mathbb{R}$ ) =>
  (Finset.range (FalihatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm) + 1)).sup'
  FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch._proof_1 fun (round :  $\mathbb{N}$ ) =>
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
    (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta (delta / 4) round)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] →  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$ 

value:
fun {Arm : Type u_1} [Fintype Arm] (lower upper delta :  $\mathbb{R}$ ) =>
  (Finset.range (FalihatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm) + 1)).sup'
  FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch._proof_1 fun (round :  $\mathbb{N}$ ) =>
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
    (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[DecidableEq Arm] →
ℕ →
(ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) →
ℕ → Finset Arm → Outcome → Finset Arm

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [DecidableEq Arm] (cutoff : ℕ)
  (decision : ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) (a : ℕ)
  (a_1 : Finset Arm) (a_2 : Outcome) =>
  if a_1.card ≤ 2 * cutoff then a_1 else FalahatgarEtAl2017MaxingRanking.pruneRound a_1 (decision a a_1 a_2)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.pruneRound](#)

FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[DecidableEq Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
Finset Arm →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome →
ℕ →
(ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) →
ℕ → PMF (Finset Arm)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [DecidableEq Arm] [Fintype Outcome] [DecidableEq Outcome]
  (initial : Finset Arm) (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome)
  (cutoff : ℕ) (decision : ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision)
  (roundCount : ℕ) =>
  FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw (PMF.pure initial) outcomeLaw
  (FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance cutoff decision) roundCount
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeSourceLogFinalOutputLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(cutoff : ℕ) →
(anchorDelta pruneDelta : ℝ) →
ℝ →
ℝ →
(lower upper : ℝ) →
ℝ →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
0 < cutoff →
0 < anchorDelta →
anchorDelta ≤ 1 →
(∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper)
```

```

      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
      maxBatch) →
    PMF Arm
value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (cutoff : ℕ)
  (anchorDelta pruneDelta fallbackDelta finalDelta lower upper epsilon : ℝ) (maxBatch : ℕ)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
  (hcutoff : 0 < cutoff) (hanchorDelta : 0 < anchorDelta) (hanchorDeltaLeOne : anchorDelta ≤ 1)
  (hbudget :
    ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
        (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
        maxBatch) =>
  PMF.map Prod.snd
  (FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackFinalLaw
    (FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw cutoff anchorDelta lower preferenceGap
      hprobability hcutoff hanchorDelta hanchorDeltaLeOne)
    (fun (anchor : Arm) =>
      FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw Finset.univ
        (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
          (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) anchor lower upper pruneDelta
            maxBatch hbudget)
          cutoff
          (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision
            (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) lower upper pruneDelta maxBatch
              hbudget)
            (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))))
    (fun (anchor : Arm) (candidates : Finset Arm) =>
      FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw anchor candidates preferenceGap
        hprobability epsilon (fallbackDelta / ↑candidates.card))
    (fun (anchor : Arm) (candidates : Finset Arm) (fallback : Arm) =>
      FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw anchor fallback candidates preferenceGap
        hprobability upper epsilon (finalDelta / ↑(Fintype.card Arm))))

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneFallbackFinalLaw](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → [DecidableEq Arm] → (roundCount : ℕ) → ℕ → ℝ → ℝ → ℝ → (Fin roundCount → Finset Arm) → ℕ
value:
fun {Arm : Type u_1} [DecidableEq Arm] (roundCount cutoff : ℕ) (lower upper delta : ℝ)
  (trace : Fin roundCount → Finset Arm) =>
  ∑ round : Fin roundCount,
    if (trace round).card ≤ 2 * cutoff then 0
    else
      (trace round).card *
        FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta ↑round)

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState

Declaration location: paper-local

Lean declaration body

```

type:
Type u_1 → ℕ → Type u_1

value:
fun (Arm : Type u_1) (roundCount : ℕ) => Finset Arm × (Fin roundCount → Finset Arm)

```

FalahatgarEtAl2017MaxingRanking.initialStoppedPruneTraceState

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[DecidableEq Arm] →
(roundCount : ℕ) → Finset Arm → FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount

value:
fun {Arm : Type u_1} [DecidableEq Arm] (roundCount : ℕ) (initial : Finset Arm) =>
(initial, fun (x : Fin roundCount) => ∅)

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → [DecidableEq Arm] → ℕ → Type (max u_1 0)

value:
fun {Arm : Type u_1} [DecidableEq Arm] (roundCount : ℕ) =>
(traceFallback : (Arm × FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool) × Arm) ×
FalahatgarEtAl2017MaxingRanking.finalCheckTraceState traceFallback.1.2.1.1

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceLaw

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
(roundCount : ℕ) →
PMF Arm →
(Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)) →
PMF (Arm × FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (roundCount : ℕ) (anchorLaw : PMF Arm)
(pruneTraceLaw : Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)) =>
anchorLaw.bind fun (anchor : Arm) =>
PMF.map
(fun (stateFailure : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool) =>
(anchor, stateFailure))
(pruneTraceLaw anchor)

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[DecidableEq Arm] →
(roundCount : ℕ) →
PMF Arm →
(Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)) →
(Arm → Finset Arm → PMF Arm) →
PMF ((Arm × FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool) × Arm)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (roundCount : ℕ) (anchorLaw : PMF Arm)
(pruneTraceLaw : Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool))
(fallbackLaw : Arm → Finset Arm → PMF Arm) =>
(FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceLaw roundCount anchorLaw pruneTraceLaw).bind
fun (anchorStateFailure : Arm × FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool) =>
PMF.map (fun (fallback : Arm) => (anchorStateFailure, fallback))
(fallbackLaw anchorStateFailure.1 anchorStateFailure.2.1.1)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[inst : DecidableEq Arm] →
(roundCount : ℕ) →
PMF Arm →
(Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)) →
(Arm → Finset Arm → PMF Arm) →
(Arm →
(candidates : Finset Arm) →
Arm → PMF (FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates)) →
PMF (FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState roundCount)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (roundCount : ℕ) (anchorLaw : PMF Arm)
(pruneTraceLaw : Arm → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool))
(fallbackLaw : Arm → Finset Arm → PMF Arm)
(finalTraceLaw :
Arm → (candidates : Finset Arm) → Arm → PMF (FalahatgarEtAl2017MaxingRanking.finalCheckTraceState candidates)) =>
(FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackLaw roundCount anchorLaw pruneTraceLaw
fallbackLaw).bind
fun (traceFallback : (Arm × FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool) × Arm) =>
PMF.map
(fun (finalTraceState : FalahatgarEtAl2017MaxingRanking.finalCheckTraceState traceFallback.1.2.1.1) =>
(traceFallback, finalTraceState))
(finalTraceLaw traceFallback.1.1 traceFallback.1.2.1.1 traceFallback.2)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackLaw](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.optMaximizeTraceFallbackFinalTraceComparisonCount

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[inst_1 : DecidableEq Arm] →
  ℕ →
  ℝ →
  ℝ →
  ℝ →
  ℝ →
  ℝ →
  ℝ →
  FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState
  (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) →
  ℕ

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (cutoff : ℕ)
  (anchorDelta pruneDelta lower upper finalEta fallbackEpsilon fallbackDelta : ℝ)
  (traceFallbackFinalTrace :
    FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState
    (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))) =>
(FalahatgarEtAl2017MaxingRanking.optMaximizePickAnchorComparisonCap cutoff anchorDelta lower +
  FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount
  (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) cutoff lower upper pruneDelta
  traceFallbackFinalTrace.fst.1.2.1.2 +
  if traceFallbackFinalTrace.snd.2 = true then 0
  else
    FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap traceFallbackFinalTrace.fst.1.2.1.1
    fallbackEpsilon fallbackDelta) +
  FalahatgarEtAl2017MaxingRanking.finalCheckTraceComparisonCount traceFallbackFinalTrace.fst.1.2.1.1 lower upper
  finalEta traceFallbackFinalTrace.snd.1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeFallbackComparisonCap](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizePickAnchorComparisonCap](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceAdvance

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[DecidableEq Arm] →
(roundCount : ℕ) →
  ℕ →
  (ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) →
  ℕ →
  FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount →
  Outcome → FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [DecidableEq Arm] (roundCount cutoff : ℕ)
  (decision : ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) (a : ℕ)
  (a_1 : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount) (a_2 : Outcome) =>
  have next : Finset Arm := FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance cutoff decision a a_1.1 a_2;
  if hround : a < roundCount then (next, Function.update a_1.2 (a, hround) a_1.1) else (next, a_1.2)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[DecidableEq Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
ℕ →
Finset Arm →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome →
(ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) →
(roundCount : ℕ) → PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [DecidableEq Arm] [Fintype Outcome] [DecidableEq Outcome]
(cutoff : ℕ) (initial : Finset Arm)
(outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome)
(decision : ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) (roundCount : ℕ) =>
FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw
(PMF.pure (FalahatgarEtAl2017MaxingRanking.initialStoppedPruneTraceState roundCount initial))
(fun (round : ℕ) (state : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount) =>
outcomeLaw round state.1)
(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceAdvance roundCount cutoff decision) roundCount
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveQueryActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.initialStoppedPruneTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(cutoff : ℕ) →
(anchorDelta pruneDelta lower upper : ℝ) →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
Arm →
0 < cutoff →
0 < anchorDelta →
anchorDelta ≤ 1 →
(∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
maxBatch) →
PMF
(Arm ×
FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) ×
Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (cutoff : ℕ)
(anchorDelta pruneDelta lower upper : ℝ) (maxBatch : ℕ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (maximum : Arm)
(hcutoff : 0 < cutoff) (hanchorDelta : 0 < anchorDelta) (hanchorDeltaLeOne : anchorDelta ≤ 1)
(hbudget :
∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
maxBatch) =>
FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceLaw
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw cutoff anchorDelta lower preferenceGap
```



```

hprobability hcutoff hanchorDelta hanchorDeltaLeOne)
fun (anchor : Arm) =>
FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw preferenceGap lower
(FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor (Fintype.card Arm) cutoff pruneDelta) cutoff
anchor Finset.univ
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) anchor lower upper pruneDelta
maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) lower upper pruneDelta maxBatch
hbudget)
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceSuccessProbability

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(cutoff : ℕ) →
(anchorDelta pruneDelta lower upper : ℝ) →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
Arm →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
0 < cutoff →
0 < anchorDelta →
anchorDelta ≤ 1 →
(∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
maxBatch) →
ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (cutoff : ℕ)
(anchorDelta pruneDelta lower upper : ℝ) (maxBatch : ℕ) (preferenceGap : Arm → Arm → ℝ) (maximum : Arm)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hcutoff : 0 < cutoff) (hanchorDelta : 0 < anchorDelta) (hanchorDeltaLeOne : anchorDelta ≤ 1)
(hbudget :
∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
maxBatch) =>
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceLaw cutoff anchorDelta pruneDelta lower
upper maxBatch preferenceGap hprobability maximum hcutoff hanchorDelta hanchorDeltaLeOne hbudget)
fun
(anchorStateFailure :
Arm ×
FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) ×
Bool) =>
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchorStateFailure.1 ∧
anchorStateFailure.2.2 = false ∧

```

```

anchorStateFailure.2.1.1.card ≤ 2 * cutoff ∧
↑(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount
  (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) cutoff lower upper
  pruneDelta anchorStateFailure.2.1.2) ≤
FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound (Fintype.card Arm) lower upper
pruneDelta ∧
(FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap upper anchorStateFailure.1 v
maximum ∈ anchorStateFailure.2.1.1)

```

Direct retained dependencies

- [AppliedModelingLib.pmfProb](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTraceFallbackFinalTraceSourceLaw

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[inst_2 : DecidableEq Arm] →
(cutoff : ℕ) →
(anchorDelta pruneDelta : ℝ) →
ℝ →
ℝ →
(lower upper : ℝ) →
ℝ →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
Arm →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
0 < cutoff →
0 < anchorDelta →
anchorDelta ≤ 1 →
(∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
maxBatch) →
PMF
(FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)))

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (cutoff : ℕ)
(anchorDelta pruneDelta fallbackDelta finalDelta lower upper epsilon : ℝ) (maxBatch : ℕ)
(preferenceGap : Arm → Arm → ℝ) (maximum : Arm)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hcutoff : 0 < cutoff) (hanchorDelta : 0 < anchorDelta) (hanchorDeltaLeOne : anchorDelta ≤ 1)
(hbudget :
  ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta pruneDelta round) ≤
  maxBatch) =>
FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceLaw
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw cutoff anchorDelta lower preferenceGap
hprobability hcutoff hanchorDelta hanchorDeltaLeOne)
(fun (anchor : Arm) =>
FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw preferenceGap lower
(FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor (Fintype.card Arm) cutoff pruneDelta) cutoff
anchor Finset.univ
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) anchor lower upper pruneDelta
maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision

```

```

(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) lower upper pruneDelta maxBatch
hbudget)
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)))
(fun (anchor : Arm) (candidates : Finset Arm) =>
  FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw anchor candidates preferenceGap
  hprobability epsilon (fallbackDelta / ↑ candidates.card))
fun (anchor : Arm) (candidates : Finset Arm) (fallback : Arm) =>
  FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceLaw anchor candidates preferenceGap hprobability upper
  epsilon (finalDelta / ↑ (Fintype.card Arm))

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinsetSeqEliminateOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorSourceOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(lower upper : ℝ) →
ℝ →
(delta : ℝ) →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
Arm →
  FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
  0 < delta →
  delta ≤ 1 →
  (∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
    (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta (delta / 4) round) ≤
    maxBatch) →
  ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (lower upper epsilon delta : ℝ) (maxBatch : ℕ)
  (preferenceGap : Arm → Arm → ℝ) (maximum : Arm)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (hdelta : 0 < delta)
  (hdeltaLeOne : delta ≤ 1)
  (hbudget :
    ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta (delta / 4) round) ≤
      maxBatch) =>
  let cutoff : ℕ := FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff (Fintype.card Arm);
  if h : delta ≤ 1 / ↑ (Fintype.card Arm) then
    FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceJointResourceProbability preferenceGap hprobability
    epsilon delta
  else
    AppliedModelingLib.pmfProb
    (FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTraceFallbackFinalTraceSourceLaw cutoff (delta / 4)
      (delta / 4) (delta / 4) (delta / 4) lower upper epsilon maxBatch preferenceGap maximum hprobability)

```

```

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability._proof_1
(FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability._proof_2
  delta hdelta)
(FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability._proof_3
  delta hdeltaLeOne)
hbudget)
fun
  (traceFallbackFinalTrace :
    FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState
      (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm))) =>
FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap epsilon
  (if traceFallbackFinalTrace.snd.2 = true then traceFallbackFinalTrace.fst.1.1
    else traceFallbackFinalTrace.fst.2) ∧
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff traceFallbackFinalTrace.fst.1.1 ∧
  traceFallbackFinalTrace.fst.1.2.2 = false ∧
  traceFallbackFinalTrace.fst.1.2.1.1.card ≤ 2 * cutoff ∧
  (FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap upper traceFallbackFinalTrace.fst.1.1 v
    maximum ∈ traceFallbackFinalTrace.fst.1.2.1.1) ∧
  ↑ (FalahatgarEtAl2017MaxingRanking.optMaximizeTraceFallbackFinalTraceComparisonCount cutoff (delta / 4)
    (delta / 4) lower upper (delta / 4 / ↑ (Fintype.card Arm)) epsilon (delta / 4)
    traceFallbackFinalTrace) ≤
  FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope (Fintype.card Arm) lower upper
  epsilon delta

```

Direct retained dependencies

- [AppliedModelingLib.pmfProb](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceJointResourceProbability](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTraceFallbackFinalTraceSourceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.finalCheckTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeAnchorPruneTraceFallbackFinalTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeTraceFallbackFinalTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivityPairFirst

Declaration location: paper-local

Lean declaration body

```

type:
{Arm : Type u_1} → [inst : DecidableEq Arm] → FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm → Arm
value:
fun {Arm : Type u_1} [DecidableEq Arm] (pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm) =>
  (Quot.out ↑ pair).1

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivityPairOfNe

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : DecidableEq Arm] →
(first second : Arm) → first ≠ second → FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (first second : Arm) (hne : first ≠ second) =>
(s(first, second), FalahatgarEtAl2017MaxingRanking.strongTransitivityPairOfNe._proof_1 first second hne)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivityPairSecond

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [inst : DecidableEq Arm] → FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm → Arm

value:
fun {Arm : Type u_1} [DecidableEq Arm] (pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm) =>
(Quot.out ↑pair).2
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingActiveSet

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] → [Nonempty Arm] → [DecidableEq Arm] → (Arm → Arm → ℝ) → ℝ → ℕ → Finset Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (estimate : Arm → Arm → ℝ) (epsilon : ℝ) (a : ℕ) =>
Nat.brecOn a (FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingActiveSet._f estimate epsilon)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.selectEstimatedThresholdCandidate](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℕ

value:
fun (armCount : ℕ) (epsilon delta : ℝ) =>
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon (delta / ↑armCount ^ 2)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[inst_1 : DecidableEq Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
(epsilon delta : ℝ) →
PMF
(FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) →
Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ) =>
let batchSize : ℕ := FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta;
let marginal :
FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm → MeasureTheory.Measure (Fin batchSize → Bool) :=
fun (pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm) =>
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.strongTransitivityPairFirst pair)
(FalahatgarEtAl2017MaxingRanking.strongTransitivityPairSecond pair) batchSize).toMeasure;
(MeasureTheory.Measure.pi marginal).toPMF
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairFirst](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairSecond](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

FalahatgarEtAl2017MaxingRanking.strongTransitivityComparisonCount

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} → [Fintype Arm] → [DecidableEq Arm] → ℝ → ℝ → ℕ

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (epsilon delta : ℝ) =>
Fintype.card (FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm) *
FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityPairEmpiricalGap

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[inst_1 : DecidableEq Arm] →
(epsilon delta : ℝ) →
(FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) →
Bool) →
FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm → ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (epsilon delta : ℝ)
(batchTable :
FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) → Bool)
(pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm) =>
FalahatgarEtAl2017MaxingRanking.strongTransitivityPairEmpiricalGap epsilon delta pair (batchTable pair)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairEmpiricalGap](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveBinaryTraceLaw

Declaration location: paper-local

Lean declaration body

```
type:
{comparisonBudget : ℕ} →
  ((round : Fin comparisonBudget) → (Fin ↑round → Bool) → PMF Bool) →
    (roundCount : ℕ) → roundCount ≤ comparisonBudget → PMF (Fin roundCount → Bool)

value:
fun {comparisonBudget : ℕ} (response : (round : Fin comparisonBudget) → (Fin ↑round → Bool) → PMF Bool) (roundCount : ℕ)
  (hroundCount : roundCount ≤ comparisonBudget) =>
  Nat.brecOn (motive := fun (x : ℕ) => x ≤ comparisonBudget → PMF (Fin x → Bool)) roundCount
    (FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveBinaryTraceLaw._f response) hroundCount
```

FalahatgarEtAl2017MaxingRanking.theorem7FirstPosition

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → 2 ≤ n → Fin n

value:
fun (n : ℕ) (hcard : 2 ≤ n) => ⟨0, FalahatgarEtAl2017MaxingRanking.theorem7FirstPosition._proof_2 n hcard⟩
```

FalahatgarEtAl2017MaxingRanking.theorem7ForwardGap

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → ℝ → Fin n → Fin n → ℝ

value:
fun (n : ℕ) (mu : ℝ) (i j : Fin n) => if ↑i = 0 ∧ ↑j + 1 = n then 1 / 2 else mu
```

FalahatgarEtAl2017MaxingRanking.theorem7LastPosition

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → 2 ≤ n → Fin n

value:
fun (n : ℕ) (hcard : 2 ≤ n) => ⟨n - 1, FalahatgarEtAl2017MaxingRanking.theorem7LastPosition._proof_2 n hcard⟩
```

FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuOutcomeProbability

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → Bool → Bool → Bool → ℝ

value:
fun (mu : ℝ) (special ordinaryDirection outcome : Bool) =>
  if special = true then if outcome = true then 1 else 0
  else if outcome = ordinaryDirection then 1 / 2 + mu else 1 / 2 - mu
```

FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuResponseLaw

Declaration location: paper-local

Lean declaration body

```

type:
(mu : ℝ) → 0 ≤ mu → mu ≤ 1 / 2 → Bool → Bool → PMF Bool

value:
fun (mu : ℝ) (hmu : 0 ≤ mu) (hmuHalf : mu ≤ 1 / 2) (special ordinaryDirection : Bool) =>
  PMF.bernoulli
    (FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuOutcomeProbability mu special ordinaryDirection true,
     FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuOutcomeProbability_nonneg† hmu hmuHalf special
      ordinaryDirection true)
    (FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuOutcomeProbability_true_le_one† hmu hmuHalf special
     ordinaryDirection)

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuOutcomeProbability](#)

FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveOrientedNoiseLaw

Declaration location: paper-local

Lean declaration body

```

type:
{comparisonBudget : ℕ} →
(mu : ℝ) →
0 ≤ mu →
mu ≤ 1 / 2 →
((round : Fin comparisonBudget) → (Fin ↑ round → Bool) → Bool) →
((round : Fin comparisonBudget) → (Fin ↑ round → Bool) → Bool) → PMF (Fin comparisonBudget → Bool)

value:
fun {comparisonBudget : ℕ} (mu : ℝ) (hmu : 0 ≤ mu) (hmuHalf : mu ≤ 1 / 2)
  (special ordinaryDirection : (round : Fin comparisonBudget) → (Fin ↑ round → Bool) → Bool) =>
  FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveBinaryTraceLaw
    (fun (round : Fin comparisonBudget) (history : Fin ↑ round → Bool) =>
      if special round history = true then AppliedModelingLib.uniformPMF Bool
      else
        FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuResponseLaw mu hmu hmuHalf false
          (ordinaryDirection round history))
    comparisonBudget (le_refl comparisonBudget)

```

Direct retained dependencies

- [AppliedModelingLib.uniformPMF](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveBinaryTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7OrientedSmallMuResponseLaw](#)

FalahatgarEtAl2017MaxingRanking.theorem7PreferenceGap

Declaration location: paper-local

Lean declaration body

```

type:
(n : ℕ) → ℝ → Fin n → Fin n → ℝ

value:
fun (n : ℕ) (mu : ℝ) (i j : Fin n) =>
  if ↑i < ↑j then FalahatgarEtAl2017MaxingRanking.theorem7ForwardGap n mu i j
  else if ↑j < ↑i then -FalahatgarEtAl2017MaxingRanking.theorem7ForwardGap n mu j i else 0

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7ForwardGap](#)

FalahatgarEtAl2017MaxingRanking.theorem7RelabeledPreferenceGap

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → ℝ → Equiv.Perm (Fin n) → Fin n → Fin n → ℝ

value:
fun (n : ℕ) (mu : ℝ) (relabel : Equiv.Perm (Fin n)) (i j : Fin n) =>
  FalahatgarEtAl2017MaxingRanking.theorem7PreferenceGap n mu ((Equiv.symm relabel) i) ((Equiv.symm relabel) j)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7PreferenceGap](#)

FalahatgarEtAl2017MaxingRanking.theorem7RelabelingOfOrderedEndpoints

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → 2 ≤ n → Fin n → Fin n → Equiv.Perm (Fin n)

value:
fun (n : ℕ) (hcard : 2 ≤ n) (first last : Fin n) =>
  have initial : Equiv.Perm (Fin n) := Equiv.swap (FalahatgarEtAl2017MaxingRanking.theorem7FirstPosition n hcard) first;
  Equiv.swap (initial (FalahatgarEtAl2017MaxingRanking.theorem7LastPosition n hcard)) last * initial
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7FirstPosition](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7LastPosition](#)

FalahatgarEtAl2017MaxingRanking.theorem7OrderedEndpointPreferenceGap

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → ℝ → 2 ≤ n → Fin n → Fin n → Fin n → Fin n → ℝ

value:
fun (n : ℕ) (mu : ℝ) (hcard : 2 ≤ n) (first last a a_1 : Fin n) =>
  FalahatgarEtAl2017MaxingRanking.theorem7RelabeledPreferenceGap n mu
  (FalahatgarEtAl2017MaxingRanking.theorem7RelabelingOfOrderedEndpoints n hcard first last) a a_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7RelabeledPreferenceGap](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7RelabelingOfOrderedEndpoints](#)

FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type (max 0 u_1)

value:
fun (Arm : Type u_1) => { pair : Sym2 Arm // ¬pair.IsDiag }
```

FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure

Declaration location: paper-local

Lean declaration body

```
field query:
{n comparisonBudget : ℕ} →
  FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
    (round : Fin comparisonBudget) →
      (Fin ↑ round → Bool) → FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)

field output:
{n comparisonBudget : ℕ} →
  FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
    (Fin comparisonBudget → Bool) → Fin (Fintype.card (Fin n)) → Fin n
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
  FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
    FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
      Bool → (roundCount : ℕ) → roundCount ≤ comparisonBudget → (Fin roundCount → Bool) → Fin roundCount → Bool

value:
fun {n comparisonBudget : ℕ}
  (procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (roundCount : ℕ) (hroundCount : roundCount ≤ comparisonBudget) (noiseHistory : Fin roundCount → Bool)
  (a : Fin roundCount) =>
  Nat.brecOn (motive := fun (x : ℕ) => x ≤ comparisonBudget → (Fin x → Bool) → Fin x → Bool) roundCount
  (FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory._f procedure coordinate orientation)
  hroundCount noiseHistory a
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomePrefix

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
  FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
    FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
      Bool → (Fin comparisonBudget → Bool) → (roundCount : ℕ) → Fin roundCount → Bool

value:
fun {n comparisonBudget : ℕ}
  (procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (noise : Fin comparisonBudget → Bool) (roundCount : ℕ) (a : Fin roundCount) =>
  Nat.brecOn (motive := fun (x : ℕ) => Fin x → Bool) roundCount
  (FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomePrefix._f procedure coordinate orientation noise) a
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeTrace

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
  FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
  FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
  Bool → (Fin comparisonBudget → Bool) → Fin comparisonBudget → Bool

value:
fun {n comparisonBudget : ℕ}
  (procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (noise : Fin comparisonBudget → Bool) (a : Fin comparisonBudget) =>
  FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomePrefix procedure coordinate orientation noise comparisonBudget a
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomePrefix](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Fin n × Fin n

value:
fun {n : ℕ} (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) =>
  let arms : Finset (Fin n) := (↑ coordinate).toFinset;
  have hcard : arms.card = 2 := FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints._proof_2 coordinate;
  have hnonempty : arms.Nonempty :=
    FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints._proof_3 coordinate hcard;
  (arms.min' hnonempty, arms.max' hnonempty)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7EndpointsOfCoordinate

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Bool → Fin n × Fin n

value:
fun {n : ℕ} (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool) =>
  if orientation = true then FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints.coordinate
  else (FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints.coordinate).swap
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinatePreferenceGap

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) → ℝ → 2 ≤ n → FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Bool → Fin n → Fin n → ℝ

value:
fun (n : ℕ) (mu : ℝ) (hcard : 2 ≤ n)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (a a_1 : Fin n) =>
  FalahatgarEtAl2017MaxingRanking.theorem7OrderedEndpointPreferenceGap n mu hcard
  (FalahatgarEtAl2017MaxingRanking.theorem7EndpointsOfCoordinate coordinate orientation).1
  (FalahatgarEtAl2017MaxingRanking.theorem7EndpointsOfCoordinate coordinate orientation).2 a a_1
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7EndpointsOfCoordinate](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7OrderedEndpointPreferenceGap](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinateOrdinaryDirection

Declaration location: paper-local

Lean declaration body

```
type:
(n : ℕ) →
ℝ →
2 ≤ n →
FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
Bool → FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Bool

value:
fun (n : ℕ) (mu : ℝ) (hcard : 2 ≤ n)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (query : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) =>
  decide
  (FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinatePreferenceGap n mu hcard coordinate orientation
   (FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints query).1
   (FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints query).2 =
   mu)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.theorem7CanonicalEndpoints](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinatePreferenceGap](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7ProcedureNoiseHistorySpecial

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
Bool → (round : Fin comparisonBudget) → (Fin ↑ round → Bool) → Bool

value:
fun {n comparisonBudget : ℕ}
  (procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
  (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool)
  (round : Fin comparisonBudget) (noiseHistory : Fin ↑ round → Bool) =>
  decide
  (procedure.query round
   (FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory procedure coordinate orientation (↑ round)
    (FalahatgarEtAl2017MaxingRanking.theorem7ProcedureNoiseHistorySpecial._proof_1 round) noiseHistory) =
   coordinate)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7ProcedureSmallMuNoiseLaw

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
(mu : ℝ) →
0 ≤ mu →
mu ≤ 1 / 2 →
2 ≤ n →
FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) →
Bool → PMF (Fin comparisonBudget → Bool)

value:
fun {n comparisonBudget : ℕ} (mu : ℝ) (hmu : 0 ≤ mu) (hmuHalf : mu ≤ 1 / 2) (hcard : 2 ≤ n)
(procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
(coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool) =>
FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveOrientedNoiseLaw mu hmu hmuHalf
(FalahatgarEtAl2017MaxingRanking.theorem7ProcedureNoiseHistorySpecial procedure coordinate orientation)
fun (round : Fin comparisonBudget) (noiseHistory : Fin ↑round → Bool) =>
FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinateOrdinaryDirection n mu hcard coordinate orientation
(procedure.query round
(FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory procedure coordinate orientation (↑round)
(FalahatgarEtAl2017MaxingRanking.theorem7ProcedureNoiseHistorySpecial._proof_1 round) noiseHistory))
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeFromNoiseHistory](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7AdaptiveOrientedNoiseLaw](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinateOrdinaryDirection](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ProcedureNoiseHistorySpecial](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7RankingSuccessAtMu

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
ℝ →
FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
2 ≤ n →
((Fin comparisonBudget → Bool) × FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) × Bool →
Prop

value:
fun {n comparisonBudget : ℕ} (mu : ℝ)
(procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget) (hcard : 2 ≤ n)
(a :
((Fin comparisonBudget → Bool) × FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) × Bool) =>
FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking
(FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinatePreferenceGap n mu hcard a.1.2 a.2) (1 / 4)
(procedure.output (FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeTrace procedure a.1.2 a.2 a.1.1))
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking](#)
- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ActualOutcomeTrace](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7HiddenCoordinatePreferenceGap](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7SmallMuInstanceSuccessProbability

Declaration location: paper-local

Lean declaration body

```
type:
{n comparisonBudget : ℕ} →
(mu : ℝ) →
0 ≤ mu →
mu ≤ 1 / 2 →
2 ≤ n →
FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget →
FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Bool → ℝ

value:
fun {n comparisonBudget : ℕ} (mu : ℝ) (hmu : 0 ≤ mu) (hmuHalf : mu ≤ 1 / 2) (hcard : 2 ≤ n)
(procedure : FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
(coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool) =>
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.theorem7ProcedureSmallMuNoiseLaw mu hmu hmuHalf hcard procedure coordinate
orientation)
fun (noise : Fin comparisonBudget → Bool) =>
FalahatgarEtAl2017MaxingRanking.theorem7RankingSuccessAtMu mu procedure hcard ((noise, coordinate), orientation)
```

Direct retained dependencies

- [AppliedModelingLib.pmfProb](#)
- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7ProcedureSmallMuNoiseLaw](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7RankingSuccessAtMu](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

FalahatgarEtAl2017MaxingRanking.theorem7SeededSmallMuInstanceSuccessProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Seed : Type u. 1} →
[Fintype Seed] →
[DecidableEq Seed] →
{n comparisonBudget : ℕ} →
PMF Seed →
(mu : ℝ) →
0 ≤ mu →
mu ≤ 1 / 2 →
2 ≤ n →
(Seed → FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget) →
FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n) → Bool → ℝ

value:
fun {Seed : Type u. 1} [Fintype Seed] [DecidableEq Seed] {n comparisonBudget : ℕ} (seedLaw : PMF Seed) (mu : ℝ)
(hmu : 0 ≤ mu) (hmuHalf : mu ≤ 1 / 2) (hcard : 2 ≤ n)
(procedure : Seed → FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget)
(coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool) =>
AppliedModelingLib.pmfExp seedLaw fun (seed : Seed) =>
FalahatgarEtAl2017MaxingRanking.theorem7SmallMuInstanceSuccessProbability mu hmu hmuHalf hcard (procedure seed)
coordinate orientation
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)
- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7SmallMuInstanceSuccessProbability](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure

Declaration location: reusable library

Lean declaration body

```
field pull:
{Arm : Type u_1} →
{pullBudget : ℕ} →
  ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget →
  (round : Fin pullBudget) → (Fin ↑ round → Bool) → Arm

field output:
{Arm : Type u_1} →
{pullBudget : ℕ} →
  ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget →
  (Fin pullBudget → Bool) → Arm
```

FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm

Declaration location: paper-local

Lean declaration body

```
field pullBudget:
{Arm : Type u_1} → FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm → ℕ

field procedure:
{Arm : Type u_1} →
  (self : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm) →
  ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm self.pullBudget

field executionComparisonCount:
{Arm : Type u_1} →
  (self : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm) → (Fin self.pullBudget → Bool) → ℕ

field executionComparisonCount_le_pullBudget:
∀ {Arm : Type u_1} (self : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm)
  (rewards : Fin self.pullBudget → Bool), self.executionComparisonCount rewards ≤ self.pullBudget

field executionComparisonCount_stable:
∀ {Arm : Type u_1} (self : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm)
  (rewards alternative : Fin self.pullBudget → Bool),
  (∀ (round : Fin self.pullBudget),
    ↑ round < self.executionComparisonCount rewards → alternative round = rewards round) →
  self.executionComparisonCount alternative = self.executionComparisonCount rewards

field output_stable_through_execution:
∀ {Arm : Type u_1} (self : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm)
  (rewards alternative : Fin self.pullBudget → Bool),
  (∀ (round : Fin self.pullBudget),
    ↑ round < self.executionComparisonCount rewards → alternative round = rewards round) →
  self.procedure.output alternative = self.procedure.output rewards
```

Direct retained dependencies

- [ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure](#)

FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardPrefixLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
{pullBudget : ℕ} →
  (winProbability : Arm → Arm → ℝ) →
  FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability →
  ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget →
  (roundCount : ℕ) → roundCount ≤ pullBudget → PMF (Fin roundCount → Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] {pullBudget : ℕ} (winProbability : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability)
  (procedure : ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget)
  (roundCount : ℕ) (hroundCount : roundCount ≤ pullBudget) =>
  Nat.brecOn (motive := fun (x : ℕ) => x ≤ pullBudget → PMF (Fin x → Bool)) roundCount
  (FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardPrefixLaw._f winProbability hprobability procedure)
  hroundCount
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaTrialLaw](#)
- [ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure](#)

FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
{pullBudget : ℕ} →
(winProbability : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability →
ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget →
PMF (Fin pullBudget → Bool)

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] {pullBudget : ℕ} (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability)
(procedure : ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget) =>
FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardPrefixLaw winProbability hprobability procedure pullBudget
(le_refl pullBudget)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardPrefixLaw](#)
- [ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure](#)

FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaSuccessProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
{pullBudget : ℕ} →
(winProbability : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability →
ℝ → ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget → ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] {pullBudget : ℕ} (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon : ℝ)
(procedure : ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure Arm pullBudget) =>
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardLaw winProbability hprobability procedure)
fun (rewards : Fin pullBudget → Bool) =>
FalahatgarEtAl2017MaxingRanking.EpsilonBordaMaximum winProbability epsilon (procedure.output rewards)
```

Direct retained dependencies

- [AppliedModelingLib.pmfProb](#)
- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonBordaMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaRewardLaw](#)
- [ZhouChenLi2014OptimalPACMultipleArm.FiniteAdaptiveBernoulliBestArmProcedure](#)

FalahatgarEtAl2017MaxingRanking.finiteAdaptiveBordaMaxingSuccessProbability

Declaration location: paper-local

Lean declaration body

```
type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
(winProbability : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability →
ℝ → FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm → ℝ
```



```

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] (winProbability : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon : ℝ)
  (algorithm : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm) =>
  FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaSuccessProbability winProbability hprobability epsilon
algorithm.procedure

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCanonicalBordaSuccessProbability](#)

Paper-specific semantic prerequisites

FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum

Source locator: cited publication, lines 54-59

Verbatim paper-source connection

If $p_{i,j} \geq 1/2$, we say that i is preferable to j and write $i \geq j$. Element i is maximal in a model if $i \geq j$ for all $j \neq i$. And a permutation $i_1, \dots, i_n$ is a ranking if $i_i \geq i_j$ for all $i \leq j$. Observe that the first element of any ranking is always maximal. For example, for $n = 3$, $p_{1,2} = 1/2$, $p_{1,3} = 1/3$, and $p_{2,3} = 2/3$, we have $1 \geq 2$, $2 \geq 1$, $3 \geq 1$, and $2 \geq 3$. Hence 2 is the unique maximum, and 2,3,1 is the unique ranking. We seek algorithms that without knowing the underlying model, use pairwise comparisons to find a maximal element and a ranking.

Lean-expanded paper semantic target

type:
{Arm : Type u_1} → (Arm → Arm → ℝ) → Arm → Prop

value:
fun {Arm : Type u_1} (preferenceGap : Arm → Arm → ℝ) (selected : Arm) =>
 ∀ (competitor : Arm), 0 ≤ preferenceGap selected competitor

Recorded source-to-declaration screening Verdict: matches
Reason: The exact declaration and Lean expansion agree with the source maximum or ranking relation.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities

Source locator: cited publication, lines 48-51

Verbatim paper-source connection

Consider a set of n elements, without loss of generality $[n] = \{1, 2, \dots, n\}$. A probabilistic model, or model for short, is an assignment of a preference probability $p_{i,j} \in [0, 1]$ for every $i \neq j \in [n]$, reflecting the probability that i is preferred when compared with j . We assume that repeated comparisons are independent and that there are no “draws”, hence $p_{j,i} = 1 - p_{i,j}$.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \text{Prop}$

value:

$\text{fun } \{\text{Arm} : \text{Type } u_1\} \text{ (winProbability : Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \Rightarrow$

$\forall (\text{arm opponent} : \text{Arm}), 0 \leq \text{winProbability arm opponent} \wedge \text{winProbability arm opponent} \leq 1$

Recorded source-to-declaration screening Verdict: matches

Reason: The exact declaration and Lean expansion preserve the source probability-range component of the comparison model.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities

Source locator: cited publication, lines 48-51

Verbatim paper-source connection

Consider a set of n elements, without loss of generality $[n] = \{1, 2, \dots, n\}$. A probabilistic model, or model for short, is an assignment of a preference probability $p_{i,j} \in [0, 1]$ for every $i \neq j \in [n]$, reflecting the probability that i is preferred when compared with j . We assume that repeated comparisons are independent and that there are no “draws”, hence $p_{j,i} = 1 - p_{i,j}$.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \text{Prop}$

value:

$\text{fun } \{\text{Arm} : \text{Type } u_1\} \text{ (preferenceGap : Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \Rightarrow$

$\forall (\text{challenger incumbent} : \text{Arm}),$

$0 \leq 1/2 + \text{preferenceGap challenger incumbent} \wedge 1/2 + \text{preferenceGap challenger incumbent} \leq 1$

Recorded source-to-declaration screening Verdict: matches

Reason: The exact declaration and Lean expansion preserve the source probability-range component of the comparison model.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.bordaScore

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.bordaScore

Source locator: cited publication, lines 112-117

Verbatim paper-source connection

score $s(i) = n^{-1} \sum_j p_{i,j}$ is the probability that i is preferable to another, randomly selected, element. Element i is Borda maximal if $s(i) = \max_j s(j)$, and ϵ -Borda maximal if $s(i) \geq \max_j s(j) - \epsilon$. A PAC Borda-maxing algorithm outputs an ϵ -Borda maximal element WHP (with probability $\geq 1 - \delta$). Similarly, a Borda ranking is a permutation $i_1, \dots, i_n$ in such that for all $1 \leq j \leq n - 1$, $s(i_j) \geq s(i_{j+1})$. An ϵ -Borda ranking is a permutation where for all $1 \leq j \leq k \leq n$, $s(i_j) \geq s(i_k) - \epsilon$. A PAC Borda-ranking algorithm outputs an ϵ -Borda ranking WHP.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow [\text{Fintype Arm}] \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \text{Arm} \rightarrow \mathbb{R}$

value:

$\text{fun } \{\text{Arm} : \text{Type } u_1\} [\text{Fintype Arm}] (\text{winProbability} : \text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) (\text{arm} : \text{Arm}) \Rightarrow$
 $(\sum \text{opponent} : \text{Arm}, \text{winProbability arm opponent}) / \uparrow (\text{Fintype.card Arm})$

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded value averages the arm's win probability over every uniformly weighted opponent, exactly the source Borda score $s(i) = (1/n) \sum_j p(i,j)$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.EpsilonBordaMaximum

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.EpsilonBordaMaximum

Source locator: cited publication, lines 112-117

Verbatim paper-source connection

score $s(i) = n^{-1} \sum_j p_{i,j}$ is the probability that i is preferable to another, randomly selected, element. Element i is Borda maximal if $s(i) = \max_j s(j)$, and ϵ -Borda maximal if $s(i) \geq \max_j s(j) - \epsilon$. A PAC Borda-maxing algorithm outputs an ϵ -Borda maximal element WHP (with probability $\geq 1 - \delta$). Similarly, a Borda ranking is a permutation $i_1, \dots, i_n$ in such that for all $1 \leq j \leq n - 1$, $s(i_j) \geq s(i_{j+1})$. An ϵ -Borda ranking is a permutation where for all $1 \leq j \leq k \leq n$, $s(i_j) \geq s(i_k) - \epsilon$. A PAC Borda-ranking algorithm outputs an ϵ -Borda ranking WHP.

Lean-expanded paper semantic target

type:

```
{Arm : Type u_1} → [Fintype Arm] → (Arm → Arm → ℝ) → ℝ → Arm → Prop
```

value:

```
fun {Arm : Type u_1} [Fintype Arm] (winProbability : Arm → Arm → ℝ) (epsilon : ℝ) (selected : Arm) =>
  ∀ (competitor : Arm),
    FalahatgarEtAl2017MaxingRanking.bordaScore winProbability competitor - epsilon ≤
    FalahatgarEtAl2017MaxingRanking.bordaScore winProbability selected
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.bordaScore](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded predicate requires the selected arm's mean Borda score to be at least every competitor's score minus epsilon, exactly the source epsilon-Borda-maximum inequality.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking

Source locator: cited publication, lines 112-117

Verbatim paper-source connection

score $s(i) = n^{-1} \sum_j p_{i,j}$ is the probability that i is preferable to another, randomly selected, element. Element i is Borda maximal if $s(i) = \max_j s(j)$, and ϵ -Borda maximal if $s(i) \geq \max_j s(j) - \epsilon$. A PAC Borda-maxing algorithm outputs an ϵ -Borda maximal element WHP (with probability $\geq 1 - \delta$). Similarly, a Borda ranking is a permutation $i_1, \dots, i_n$ in such that for all $1 \leq j \leq n - 1$, $s(i_j) \geq s(i_{j+1})$. An ϵ -Borda ranking is a permutation where for all $1 \leq j \leq k \leq n$, $s(i_j) \geq s(i_k) - \epsilon$. A PAC Borda-ranking algorithm outputs an ϵ -Borda ranking WHP.

Lean-expanded paper semantic target

type:
{Arm : Type u_1} → [inst : Fintype Arm] → (Arm → Arm → ℝ) → ℝ → (Fin (Fintype.card Arm) → Arm) → Prop

value:
fun {Arm : Type u_1} [Fintype Arm] (winProbability : Arm → Arm → ℝ) (epsilon : ℝ)
 (ranking : Fin (Fintype.card Arm) → Arm) =>
 Function.Bijective ranking ∧
 ∀ (first second : Fin (Fintype.card Arm)),
 ↑first ≤ ↑second →
 FalahatgarEtAl2017MaxingRanking.bordaScore winProbability (ranking second) - epsilon ≤
 FalahatgarEtAl2017MaxingRanking.bordaScore winProbability (ranking first)

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.bordaScore](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded predicate requires a bijective ranking and, for every earlier-later pair, the earlier Borda score to be at least the later score minus epsilon, exactly the source epsilon-Borda-ranking definition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.EpsilonMaximum

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.EpsilonMaximum

Source locator: cited publication, lines 65-72

Verbatim paper-source connection

Let $\tilde{p}_{i,j} = p_{i,j} - 1/2$ be the centered preference probability. Note that $\tilde{p}_{i,j} \geq 0$ iff i is preferable to j . If $\tilde{p}_{i,j} \geq -\epsilon$ we say that i is ϵ -preferable to j . For $0 < \epsilon < 1/2$, an element $i \in [n]$ is ϵ -maximum if it is ϵ -preferable to all other elements, namely, $\tilde{p}_{i,j} \geq -\epsilon \forall j \neq i$. Given $\epsilon > 0$, $1/2 \geq \delta > 0$, a PAC maxing algorithm must output an ϵ -maxima with probability $\geq 1 - \delta$, henceforth abbreviated with high probability (WHP). Similarly, a permutation π of $\{1, \dots, n\}$ is an ϵ -ranking if $\pi(i)$ is ϵ -preferable to $\pi(j)$ for all $i \leq j$, and a PAC ranking algorithm must output an ϵ -ranking WHP. Note that in this paper, we consider $\delta \leq 2^{-1}$, the more practical regime. For larger values of δ , one can use our algorithms with $\delta = 1/2$.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \text{Arm} \rightarrow \text{Prop}$

value:

$\text{fun } \{\text{Arm} : \text{Type } u_1\} \text{ (preferenceGap : Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \text{ (epsilon : } \mathbb{R}) \text{ (selected : Arm)} \Rightarrow$
 $\text{AppliedModelingLib.PreferenceRL.ApproximatePreferenceWinner preferenceGap selected epsilon}$

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.ApproximatePreferenceWinner](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded predicate says the selected arm's centered preference gap against every competitor is at least minus epsilon, exactly the source epsilon-maximum definition; including the arm itself is harmless because the model fixes its self-gap at zero.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking

Source locator: cited publication, lines 65-72

Verbatim paper-source connection

Let $\tilde{p}_{i,j} = p_{i,j} - 1/2$ be the centered preference probability. Note that $\tilde{p}_{i,j} \geq 0$ iff i is preferable to j . If $\tilde{p}_{i,j} \geq -\epsilon$ we say that i is ϵ -preferable to j . For $0 < \epsilon < 1/2$, an element $i \in [n]$ is ϵ -maximum if it is ϵ -preferable to all other elements, namely, $\tilde{p}_{i,j} \geq -\epsilon \forall j \neq i$. Given $\epsilon > 0$, $1/2 \geq \delta > 0$, a PAC maxing algorithm must output an ϵ -maxima with probability $\geq 1 - \delta$, henceforth abbreviated with high probability (WHP). Similarly, a permutation π of $\{1, \dots, n\}$ is an ϵ -ranking if $\pi(i)$ is ϵ -preferable to $\pi(j)$ for all $i \leq j$, and a PAC ranking algorithm must output an ϵ -ranking WHP. Note that in this paper, we consider $\delta \leq 2^{-1}$, the more practical regime. For larger values of δ , one can use our algorithms with $\delta = 1/2$.

Lean-expanded paper semantic target

type:
 $\{\text{Arm} : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype Arm}] \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow (\text{Fin} (\text{Fintype.card Arm}) \rightarrow \text{Arm}) \rightarrow \text{Prop}$

value:
 $\text{fun } \{\text{Arm} : \text{Type } u_1\} [\text{Fintype Arm}] (\text{preferenceGap} : \text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) (\text{epsilon} : \mathbb{R})$
 $(\text{ranking} : \text{Fin} (\text{Fintype.card Arm}) \rightarrow \text{Arm}) \Rightarrow$
 $\text{Function.Bijective ranking} \wedge$
 $\forall (\text{first second} : \text{Fin} (\text{Fintype.card Arm})),$
 $\uparrow \text{first} \leq \uparrow \text{second} \rightarrow -\text{epsilon} \leq \text{preferenceGap} (\text{ranking first}) (\text{ranking second})$

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded predicate is a bijection and imposes centered gap at least minus epsilon for every earlier-later pair, exactly the source epsilon-ranking condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.GoodAnchor

Source locator: cited publication, lines 263-264

Verbatim paper-source connection

An element a is called an (ϵ, n') -good anchor if a is not ϵ -preferable to at most n' elements, i.e.,
 $|\{e : e \in S \text{ and } \tilde{p}_{e,a} > \epsilon\}| \leq n'$.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow [\text{Fintype Arm}] \rightarrow [\text{DecidableEq Arm}] \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \text{Arm} \rightarrow \text{Prop}$

value:

```
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ) (epsilon : ℝ) (cutoff : ℕ)
  (anchor : Arm) =>
  {arm : Arm | epsilon < preferenceGap arm anchor}.card ≤ cutoff
```

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded Finset cardinality is the number of arms whose centered gap against the anchor is strictly greater than epsilon, bounded by the cutoff, exactly the source good-anchor definition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule

Source locator: cited publication, lines 1248-1256

Verbatim paper-source connection

Algorithm 9 B ORDA -R ANKING

```
1: inputs
2: set S, bias  $\epsilon$ , confidence  $\delta$ .
3: Initialize:  $be \leftarrow 21$  for all  $e \in S$ 
4: for element  $e$  in S do
5:    $be \leftarrow E\text{ STIMATE -B ORDA -S CORE}(S, e, 2\epsilon, n\delta)$ 
6: end for
7: Rank S according to  $be$ .
8: return S.
```

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype Arm}] \rightarrow ((\text{Arm} \rightarrow \mathbb{R}) \rightarrow \text{Fin} (\text{Fintype.card Arm}) \rightarrow \text{Arm}) \rightarrow \text{Prop}$

value:

```
fun {Arm : Type u_1} [Fintype Arm] (rankingRule : (Arm → ℝ) → Fin (Fintype.card Arm) → Arm) =>
  ∀ (estimate : Arm → ℝ),
    Function.Bijective (rankingRule estimate) ∧
      FalahatgarEtAl2017MaxingRanking.EstimatedScoresSorted estimate (rankingRule estimate)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.EstimatedScoresSorted](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded rule returns each arm exactly once and orders all earlier-later pairs by nonincreasing supplied empirical Borda score; this is exactly Algorithm 9's instruction to rank by the estimates, with source-unspecified tie breaking left open.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.PreferenceRanking

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.PreferenceRanking

Source locator: cited publication, lines 54-59

Verbatim paper-source connection

If $p_{i,j} \geq 1/2$, we say that i is preferable to j and write $i \geq j$. Element i is maximal in a model if $i \geq j$ for all $j \neq i$. And a permutation $\langle 1, \dots, n \rangle$ is a ranking if $i \geq j$ for all $i \leq j$. Observe that the first element of any ranking is always maximal. For example, for $n = 3$, $p_{1,2} = 1/2$, $p_{1,3} = 1/3$, and $p_{2,3} = 2/3$, we have $1 \geq 2$, $2 \geq 1$, $3 \geq 1$, and $2 \geq 3$. Hence 2 is the unique maximum, and 2,3,1 is the unique ranking. We seek algorithms that without knowing the underlying model, use pairwise comparisons to find a maximal element and a ranking.

Lean-expanded paper semantic target

type:

```
{Arm : Type u_1} → [inst : Fintype Arm] → (Arm → Arm → ℝ) → (Fin (Fintype.card Arm) → Arm) → Prop
```

value:

```
fun {Arm : Type u_1} [Fintype Arm] (preferenceGap : Arm → Arm → ℝ) (ranking : Fin (Fintype.card Arm) → Arm) =>  
  Function.Bijective ranking ∧  
  ∀ (first second : Fin (Fintype.card Arm)), ↑first ≤ ↑second → 0 ≤ preferenceGap (ranking first) (ranking second)
```

Recorded source-to-declaration screening Verdict: matches

Reason: The exact declaration and Lean expansion agree with the source maximum or ranking relation.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity

Source locator: cited publication, lines 88-93

Verbatim paper-source connection

But significantly more general, and more realistic, non-parametric, models may also have maxima and rankings. A model is strongly stochastically transitive (SST), if $i \geq j$ and $j \geq k$ imply $p_{i,k} \geq \max(p_{i,j}, p_{j,k})$. By simple induction, every SST model has a maximum element and a ranking. And one additional property, that is perhaps more difficult to justify, has proved helpful in constructing maxing and sorting PAC algorithms. A tournament satisfies the stochastic triangle inequality if $i \geq j$ and $j \geq k$ imply that $\tilde{p}_{i,k} \leq \tilde{p}_{i,j} + \tilde{p}_{j,k}$.

Lean-expanded paper semantic target

type:

$\{\text{Arm} : \text{Type } u_1\} \rightarrow (\text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \rightarrow \text{Prop}$

value:

$\text{fun } \{\text{Arm} : \text{Type } u_1\} (\text{preferenceGap} : \text{Arm} \rightarrow \text{Arm} \rightarrow \mathbb{R}) \Rightarrow$

$\forall (\text{first middle last} : \text{Arm}),$

$0 \leq \text{preferenceGap first middle} \rightarrow$

$0 \leq \text{preferenceGap middle last} \rightarrow$

$\max (\text{preferenceGap first middle}) (\text{preferenceGap middle last}) \leq \text{preferenceGap first last}$

Recorded source-to-declaration screening Verdict: matches

Reason: The exact declaration and Lean expansion preserve both preference premises and the source maximum-gap SST conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.adaptiveCompare

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.adaptiveCompare

Source locator: cited publication, lines 737-759

Verbatim paper-source connection

Algorithm 4 C OMPARE

```
1: inputs
2:   element i, element j, bias lower limit  $\epsilon_l \geq 0$ , bias upper limit  $\epsilon_u > \epsilon_l$ , confidence  $\delta$ 
3: initialize
4:    $\epsilon_m = (\epsilon_l + \epsilon_u)/2$ ,  $\hat{p}^{i,j} \leftarrow 0$ ,  $\hat{c} \leftarrow 21$ ,  $t \leftarrow 0$ ,  $w \leftarrow 0$ 
5: while  $|\hat{p}^{i,j} - \epsilon_m| \leq \frac{\hat{c}}{2}$  and  $t \leq \frac{(\epsilon_u - \epsilon_l)^2}{2 \log \delta}$  do
6:   Compare i and j
7:   if i wins then
8:      $w \leftarrow w+1$ 
9:   end if
10:   $t \leftarrow t+1$ 
11:   $\hat{p}^{i,j} \leftarrow \frac{wt}{t+1}$ ,  $\hat{c} \leftarrow 2t$ 
12: end while
13: if  $|\hat{p}^{i,j} - \epsilon_m| \leq \epsilon_m$  then
14:   return 1
15: end if
16: return 2
```

Lean-expanded paper semantic target

type:
{ Ω : Type u_1} → ($\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$) → $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \Omega \rightarrow$ FalahatgarEtAl2017MaxingRanking.CompareDecision

value:
fun { Ω : Type u_1} (observation : $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$) (count : $\mathbb{N}$) (lower upper delta : $\mathbb{R}$) (outcome : Ω) =>
if hstop :
 $\exists$ time $\in$ Finset.lcc 1 count,
 FalahatgarEtAl2017MaxingRanking.adaptiveConfidenceRadius time delta <
 |FalahatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate observation time outcome -
 FalahatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint lower upper| then
if
 FalahatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate observation (Nat.find hstop) outcome $\leq$
 FalahatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint lower upper then
 FalahatgarEtAl2017MaxingRanking.CompareDecision.lower
else FalahatgarEtAl2017MaxingRanking.CompareDecision.upper
else
if
 FalahatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate observation count outcome $\leq$
 FalahatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint lower upper then
 FalahatgarEtAl2017MaxingRanking.CompareDecision.lower
else FalahatgarEtAl2017MaxingRanking.CompareDecision.upper

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCenteredEstimate](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompareMidpoint](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveConfidenceRadius](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded function searches the source confidence-radius stopping condition through the finite cap, then compares the centered empirical estimate with the midpoint of the lower and upper biases to return the corresponding one of the two Algorithm 4 decisions.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore

Source locator: cited publication, lines 1228-1240

Verbatim paper-source connection

Algorithm 8 E STIMATE -B ORDA -S CORE

```
1: inputs
2: set S, element e, bias  $\epsilon$ , confidence  $\delta$ .
3: Initialize:  $w \leftarrow 0$ ,  $\hat{s} \leftarrow 21$ ,  $m \leftarrow 2\epsilon 12 \log 2\delta$ .
4: for  $k = 1$  to  $k = m$  do
5: Compare e with random element  $\in S$ 
6: if e wins then
7:    $w \leftarrow w+1$ 
8: end if
9:  $\hat{s} = w$ 
    $k$ 
10: end for
11: return  $\hat{s}$ 
```

Lean-expanded paper semantic target

type:

$\{\Omega : \text{Type } u_1\} \rightarrow (\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$

value:

fun $\{\Omega : \text{Type } u_1\}$ (observation : $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$) (count : $\mathbb{N}$) (outcome : Ω) =>
 $(\sum \text{index} \in \text{Finset.range count, observation index outcome}) / \uparrow \text{count}$

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded value is the number of wins in the sampled prefix divided by its length, exactly Algorithm 8's returned empirical win fraction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw

Source locator: cited publication, lines 880-884

Verbatim paper-source connection

Algorithm 5 P ICK -A NCHOR

- 1: inputs
- 2: Set S of size n , size n' , bias ϵ , confidence δ .
- 3: Form a set Q by selecting $\min(n, n' \log 2\delta, n)$ random elements from S without replacement.
- 4: return S EQ -E LIMINATE($Q, \epsilon, 2\delta$)

Lean-expanded paper semantic target

```

type:
{Arm : Type u_1} →
{count : ℕ} →
  AppliedModelingLib.finiteFreshList Arm count ∅ →
  0 < count →
    (preferenceGap : Arm → Arm → ℝ) →
      FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
        (epsilon eta : ℝ) →
          ℕ → Arm → PMF (Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon eta) → Bool)

value:
fun {Arm : Type u_1} {count : ℕ} (sample : AppliedModelingLib.finiteFreshList Arm count ∅) (hcount : 0 < count)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon eta : ℝ)
  (a : ℕ) (a_1 : Arm) =>
  FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareOutcomeLaw
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleInitial sample hcount)
  (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleChallengers sample) preferenceGap hprobability epsilon eta a a_1

```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshAdaptiveCompareOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleChallengers](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleInitial](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Algorithm 5 first samples a finite set without replacement and then runs its adaptive Seq-Eliminate calls conditional on that realized sample. This declaration is exactly that source-faithful history-selected Bernoulli comparison kernel; it adds no outcome, assumption, or algorithmic branch.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateOutputLaw

Source locator: cited publication, lines 222-233

Verbatim paper-source connection

```

Algorithm 1 S EQ -E LIMINATE
1: inputs
2: Set S, bias  $\epsilon$ , confidence  $\delta$ 
3:  $n \leftarrow |S|$ 
4:  $r \leftarrow$  a random  $c \in S$ ,  $S = S \setminus \{r\}$ 
5: while  $S \neq \emptyset$  do
6:   Pick a random  $c \in S$ ,  $S = S \setminus \{c\}$ .
7:   if  $C\_OMPARE(c, r, 0, \epsilon, n\delta) = 2$  then
8:      $r \leftarrow c$ 
9:   end if
10: end while
11: return r

```

Lean-expanded paper semantic target

```

type:
{Arm : Type u_1} →
[Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap → ℝ → ℝ → PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ) =>
PMF.map
(fun (sampleStateFailure : AppliedModelingLib.finiteFreshList Arm (Fintype.card Arm)  $\emptyset \times$  Arm  $\times$  Bool) =>
sampleStateFailure.2.1)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateJointLaw preferenceGap hprobability epsilon delta)

```

Direct retained dependencies

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateJointLaw](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Algorithm 1 starts with a random running arm and repeatedly eliminates a randomly selected challenger. This PMF forgets only the sampled full order and returns the resulting arm, preserving the source randomized Seq-Eliminate execution and comparison model.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeSourceLogAlgorithmOutputLaw

Source locator: cited publication, lines 375-391

Verbatim paper-source connection

```

Algorithm 3 OPT-MAXIMIZE
1: inputs
2:  Set  $S$ , bias  $\epsilon$ , confidence  $\delta$ .
3: if  $\delta \leq n_1$  then
4:   return  $S$  EQ-E-LIMINATE( $S$ ,  $\epsilon$ ,  $\delta$ )
5: end if
6:  $a \leftarrow \text{PICK-A-NCHOR}(S, 6n \log n, \epsilon/3, 4\delta)$ 
7:  $S' \leftarrow \text{P-RUNE}(S, a, 6n \log n, \epsilon/3, 2\epsilon/3, 4\delta)$ 
8: for element  $e$  in  $S'$  do
9:  if COMPARE( $e$ ,  $a$ ,  $2\epsilon/3$ ,  $\epsilon, 4n$ ) = 2 then
10:   return  $S$  EQ-E-LIMINATE( $S'$ ,  $\epsilon$ ,  $4\delta$ )
11: end if
12: end for
13: return  $a$ 

```

Lean-expanded paper semantic target

```

type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[Nonempty Arm] →
[DecidableEq Arm] →
(cutoff : ℕ) →
(lower upper : ℝ) →
ℝ →
(delta : ℝ) →
(maxBatch : ℕ) →
(preferenceGap : Arm → Arm → ℝ) →
FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap →
0 < delta →
delta ≤ 1 →
0 < cutoff →
(∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta (delta / 4) round) ≤
maxBatch) →
PMF Arm

value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (cutoff : ℕ) (lower upper epsilon delta : ℝ)
(maxBatch : ℕ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (hdelta : 0 < delta)
(hdeltaLeOne : delta ≤ 1) (hcutoff : 0 < cutoff)
(hbudget :
  ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta (delta / 4) round) ≤
  maxBatch) =>
if hbase : delta ≤ 1 / ↑(Fintype.card Arm) then
  FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw preferenceGap hprobability epsilon delta
else
  FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeSourceLogFinalOutputLaw cutoff (delta / 4) (delta / 4)
  (delta / 4) (delta / 4) lower upper epsilon maxBatch preferenceGap hprobability hcutoff
  (FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTraceAlgorithmSourceOutputLaw._proof_1 delta hdelta)
  (FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTraceAlgorithmSourceOutputLaw._proof_2 delta hdeltaLeOne)
  hbudget

```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeBaseSourceOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeSourceLogFinalOutputLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

Recorded source-to-declaration screening **Verdict:** matches

Reason: The two branches are Algorithm 3's printed Seq-Eliminate fallback and its Pick-Anchor, Prune, and final-check path, with the source confidence allocations and source Prune horizon.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate

Source locator: cited publication, lines 1182-1199

Verbatim paper-source connection

Algorithm 7 S TRONG -T RANSITIVITY-R ANKING

```
1: inputs
2: Set S, bias  $\epsilon$ , confidence  $\delta$ 
3: for every pair  $\{i, j\}$  such that  $i, j \in S$  do
4:  $\hat{p}_{i,j} \leftarrow \text{E STIMATE -P ROBABILITY}(i, j, \epsilon/2, \delta/n^2)$ 
5:  $\hat{p}_{j,i} \leftarrow 1 - \hat{p}_{i,j}$ 
6: end for
7: ordered set  $T \leftarrow \emptyset$ 
8: while  $|S| > 0$  do
9: if  $\exists e$  s.t.  $\hat{p}_{e,f} \geq 12 - \epsilon \forall f \in S$  then
10:   Add e at the end of T
11:    $S = S \setminus \{e\}$ 
12: else
13:   Add S at the end of T
14:   return T
15: end if
16: end while
17: return T
```

[Next verbatim source excerpt]

Lemma 20. S TRONG -T RANSITIVITY-R ANKING(S, ϵ, δ) uses $O(n^2 \log n \delta)$ comparisons and w.p. $\geq 1 - \delta$ returns an ϵ -ranking.

Proof. S TRONG -T RANSITIVITY-R ANKING calls E STIMATE -P ROBABILITY for $O(n^2)$ times, once for each pair and each EP $(i, j, \epsilon/2, \delta/n^2)$ uses $O(\epsilon^2 \log n \delta)$ comparisons and hence bound on comparisons follow.

17

[form-feed in source extraction]

w.p. $\geq 1 - \delta/n^2$, E STIMATE -P ROBABILITY($i, j, \epsilon/2, \delta/n^2$) approximates $p_{i,j}$ with $\hat{p}_{i,j}$ such that $|p_{i,j} - \hat{p}_{i,j}| \leq \frac{\epsilon}{2}$

. By the union bound, w.p. $\geq 1 - \delta$, $|p_{i,j} - \hat{p}_{i,j}| \leq 2\epsilon \forall i, j \in S$. From here we assume that $|p_{i,j} - \hat{p}_{i,j}| \leq 2\epsilon \forall i, j \in S$ and show that the output is an ϵ -ranking. Let S_t denote the set of remaining elements in S after t elements are removed from S . We first show that for $0 \leq t \leq n - 1$, there is one element e such that $\hat{p}_{e,j} \geq 21 - \epsilon \forall j \in S_t$ and such an element is an ϵ -maximum of S_t . Observe that if there is an element e such that $\hat{p}_{e,j} \geq 21 - \epsilon/2 \forall j \in S_t$ then $p_{e,j} \geq 12 - \epsilon \forall j \in S_t$ and hence e is an ϵ -maximum of S_t . Further recall that for the absolute maximum at^* of S_t , $p_{at^*,j} \geq 21 \forall j \in S_t$ and hence $\hat{p}_{at^*,j} \geq 12 - \epsilon/2 \forall j \in S_t$. Therefore there will be at least one element e s.t. $\hat{p}_{e,j} \geq 21 - \epsilon/2$ and such an element will be an ϵ -maximum of S_t . S TRONG -T RANSITIVITY-R ANKING deletes one such element from S_t and adds it to the end of the ordered output set. Since for every t , S TRONG -T RANSITIVITY-R ANKING adds an ϵ -maximum of S_t to the output set, the Lemma follows.

Lean-expanded paper semantic target

type:

```
{Arm : Type u_1} →
[inst : Fintype Arm] →
[inst_1 : DecidableEq Arm] →
(epsilon delta : ℝ) →
(FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) →
Bool) →
Arm → Arm → ℝ
```

value:

```
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (epsilon delta : ℝ)
(batchTable :
FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) → Bool)
(first second : Arm) =>
if hsame : first = second then 0
else
have pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm :=
FalahatgarEtAl2017MaxingRanking.strongTransitivityPairOfNe first second hsame;
if first = FalahatgarEtAl2017MaxingRanking.strongTransitivityPairFirst pair then
FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityPairEmpiricalGap epsilon delta batchTable pair
else -FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityPairEmpiricalGap epsilon delta batchTable pair
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityPairEmpiricalGap](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairFirst](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityPairOfNe](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

Recorded source-to-declaration screening **Verdict:** matches

Reason: Algorithm 7 estimates each unordered pair once and defines the reverse empirical entry as its complement. The centered form here uses the sampled orientation directly, its negative for the reverse orientation, and zero on the diagonal, exactly implementing that source algorithm under the documented display correction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget

Source locator: cited publication, lines 1228-1240

Verbatim paper-source connection

Algorithm 8 E STIMATE -B ORDA -S CORE

```
1: inputs
2: set S, element e, bias  $\epsilon$ , confidence  $\delta$ .
3: Initialize:  $w \leftarrow 0$ ,  $\hat{s} \leftarrow 21$ ,  $m \leftarrow 2\epsilon 12 \log 2\delta$ .
4: for  $k = 1$  to  $k = m$  do
5: Compare e with random element  $\in S$ 
6: if e wins then
7:    $w \leftarrow w+1$ 
8: end if
9:  $\hat{s} = w$ 
    $k$ 
10: end for
11: return  $\hat{s}$ 
```

Lean-expanded paper semantic target

type:

$\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$

value:

fun (epsilon delta : $\mathbb{R}$) => FalahatgarEtAl2017MaxingRanking.meanEstimateSampleBudget epsilon delta

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.meanEstimateSampleBudget](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded natural ceiling is $\text{ceil}(\log(2/\delta)/(2 \epsilon^2))$, exactly the finite integer realization of Algorithm 8's displayed sample count.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget

Source locator: cited publication, lines 1169-1173

Verbatim paper-source connection

Algorithm 6 E STIMATE -P ROBABILITY

- 1: inputs
- 2: element i, element j, bias ϵ , confidence δ .
- 3: Compare i and j for $2\epsilon \log 2/\delta$ times.
- 4: return Fraction of times i won

Lean-expanded paper semantic target

type:

$\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N}$

value:

fun (epsilon delta : $\mathbb{R}$) => FalahatgarEtAl2017MaxingRanking.meanEstimateSampleBudget epsilon delta

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.meanEstimateSampleBudget](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded natural ceiling is $\text{ceil}(\log(2/\delta)/(2\epsilon^2))$, exactly Algorithm 6's number of pairwise comparisons.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw

Source locator: cited publication, lines 308-324

Verbatim paper-source connection

Algorithm 2 P RUNE

```
1: inputs
2: Set S, element a, size n', lower bias  $\epsilon_l$ , upper bias  $\epsilon_u$ , confidence  $\delta$ .
3:  $t \leftarrow 1$ 
4:  $S_1 \leftarrow S$ 
5: while  $|S_t| > 2n'$  and  $t < \log n$  do
6:   Initialize:  $Q_t \leftarrow \emptyset$ 
7:   for e in  $S_t$  do
8:     if  $C\_COMPARE(e, a, \epsilon_l, \epsilon_u, \delta/2t+1) = 1$  then
9:        $Q_t \leftarrow Q_t \cup \{e\}$ 
10:    end if
11:  end for
12:  $S_{t+1} \leftarrow S_t \setminus Q_t$ 
13:  $t \leftarrow t+1$ 
14: end while
15: return  $S_t$ .
```

Lean-expanded paper semantic target

```
type:
{Arm : Type u_1} →
{Outcome : Type u_2} →
[Fintype Arm] →
[DecidableEq Arm] →
[Fintype Outcome] →
[DecidableEq Outcome] →
(Arm → Arm → ℝ) →
ℝ →
ℝ →
ℕ →
Arm →
Finset Arm →
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome →
(ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) →
(roundCount : ℕ) →
PMF (FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount × Bool)

value:
fun {Arm : Type u_1} {Outcome : Type u_2} [Fintype Arm] [DecidableEq Arm] [Fintype Outcome] [DecidableEq Outcome]
  (preferenceGap : Arm → Arm → ℝ) (lower delta : ℝ) (cutoff : ℕ) (anchor : Arm) (initial : Finset Arm)
  (outcomeLaw : AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm) Outcome)
  (decision : ℕ → Finset Arm → Outcome → Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision) (roundCount : ℕ) =>
AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw
(PMF.pure (FalahatgarEtAl2017MaxingRanking.initialStoppedPruneTraceState roundCount initial))
(fun (round : ℕ) (state : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount) =>
  outcomeLaw round state.1)
(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceAdvance roundCount cutoff decision)
(fun (round : ℕ) (state : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount)
  (outcome : Outcome) =>
  →state.1.card ≤ 2 * cutoff ∧
  ↑cutoff < ↑(FalahatgarEtAl2017MaxingRanking.pruneBadArms preferenceGap lower anchor state.1).card ∧
  delta * ↑(FalahatgarEtAl2017MaxingRanking.pruneBadArms preferenceGap lower anchor state.1).card <
  ↑(FalahatgarEtAl2017MaxingRanking.pruneBadArms preferenceGap lower anchor
    (FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance cutoff decision round state.1 outcome)).card)
roundCount
```

Direct retained dependencies

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.adaptiveQueryStateLaw](#)
- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.initialStoppedPruneTraceState](#)
- [FalahatgarEtAl2017MaxingRanking.pruneBadArms](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceAdvance](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

Recorded source-to-declaration screening **Verdict:** matches

Reason: This is Algorithm 2's stopped active-set execution: it starts from S, applies the round decision while the size test permits, and records the resulting finite trace under the printed round guard.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw

Source locator: cited publication, lines 880-884

Verbatim paper-source connection

Algorithm 5 P ICK -A NCHOR

- 1: inputs
- 2: Set S of size n , size n' , bias ϵ , confidence δ .
- 3: Form a set Q by selecting $\min(nn' \log 2\delta, n)$ random elements from S without replacement.
- 4: return $S \text{ EQ } -E \text{ LIMINATE}(Q, \epsilon, 2\delta)$

Lean-expanded paper semantic target

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[DecidableEq Arm] → (count : ℕ) → count ≤ Fintype.card Arm → PMF (AppliedModelingLib.finiteFreshList Arm count ∅)
```

```
value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (count : ℕ) (hcount : count ≤ Fintype.card Arm) =>
AppliedModelingLib.finiteWithoutReplacementPMF (fun (x : Arm) => 1)
FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw._proof_1
FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw._proof_2 count ∅
(FalahatgarEtAl2017MaxingRanking.pickAnchorUniformSampleLaw._proof_3 count hcount)
```

Direct retained dependencies

- [AppliedModelingLib.finiteAvailableWeight](#)
- [AppliedModelingLib.finiteFreshList](#)
- [AppliedModelingLib.finiteWithoutReplacementPMF](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source Pick-Anchor algorithm chooses its Q sample uniformly without replacement. This finite PMF gives exactly the ordered uniform-without-replacement law, with the sample-size cap made explicit and duplicates excluded by construction.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff

Source locator: cited publication, lines 375-391

Verbatim paper-source connection

```
Algorithm 3 OPT-MAXIMIZE
1: inputs
2:   Set S, bias  $\epsilon$ , confidence  $\delta$ .
3: if  $\delta \leq n$  then
4:   return S EQ-E LIMINATE(S,  $\epsilon$ ,  $\delta$ )
5: end if
6:  $a \leftarrow \text{PICK-A-NCHOR}(S, \sqrt{6n \log n}, \epsilon/3, 4\delta)$ 
7:  $S' \leftarrow \text{PRUNE}(S, a, \sqrt{6n \log n}, \epsilon/3, 2\epsilon/3, 4\delta)$ 
8: for element  $e$  in  $S'$  do
9:   if COMPARE(e, a,  $2\epsilon/3$ ,  $\sqrt{6n \log n}, \epsilon, 4n$ ) = 2 then
10:    return S EQ-E LIMINATE( $S'$ ,  $\epsilon$ ,  $4\delta$ )
11:   end if
12: end for
13: return a
```

Lean-expanded paper semantic target

```
type:
  ℕ → ℕ

value:
  fun (armCount : ℕ) => min armCount ([√(6 * ↑armCount * Real.log ↑armCount)] + 1)
```

Recorded source-to-declaration screening Verdict: matches

Reason: This is the total integer cutoff used for Algorithm 3’s displayed square-root Prune parameter; its cap by the input size leaves the source call within its feasible active set.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.stoppedPruneRounds

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.stoppedPruneRounds

Source locator: cited publication, lines 308-324

Verbatim paper-source connection

Algorithm 2 P RUNE

```
1: inputs
2: Set S, element a, size n', lower bias  $\epsilon_l$ , upper bias  $\epsilon_u$ , confidence  $\delta$ .
3:  $t \leftarrow 1$ 
4:  $S_1 \leftarrow S$ 
5: while  $|S_t| > 2n'^2$  and  $t < \log n$  do
6:   Initialize:  $Q_t \leftarrow \emptyset$ 
7:   for e in  $S_t$  do
8:     if C OMPARE(e, a,  $\epsilon_l$ ,  $\epsilon_u$ ,  $\delta/2t+1$ ) = 1 then
9:        $Q_t \leftarrow Q_t \cup \{e\}$ 
10:    end if
11:  end for
12:  $S_{t+1} \leftarrow S_t \setminus Q_t$ 
13:  $t \leftarrow t+1$ 
14: end while
15: return  $S_t$ .
```

Lean-expanded paper semantic target

```
type:
{Ω : Type u_1} →
{Arm : Type u_2} →
[DecidableEq Arm] →
ℕ → ℕ → (ℕ → Arm → Ω → FalahatgarEtAl2017MaxingRanking.CompareDecision) → Ω → Finset Arm → Finset Arm
```

```
value:
fun {Ω : Type u_1} {Arm : Type u_2} [DecidableEq Arm] (roundCount cutoff : ℕ)
  (decision : ℕ → Arm → Ω → FalahatgarEtAl2017MaxingRanking.CompareDecision) (outcome : Ω) (active : Finset Arm) =>
  Nat.brecOn roundCount (FalahatgarEtAl2017MaxingRanking.stoppedPruneRounds._f cutoff decision outcome active)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.pruneRound](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This recursively applies Algorithm 2's remove-and-stop rule for a supplied finite horizon. Instantiated at the source squared-log count, it follows the printed one-indexed loop.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.strongTransitivityPairEmpiricalGap

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.strongTransitivityPairEmpiricalGap

Source locator: cited publication, lines 1169-1173

Verbatim paper-source connection

Algorithm 6 E STIMATE -P ROBABILITY

- 1: inputs
- 2: element i , element j , bias ϵ , confidence δ .
- 3: Compare i and j for $2\epsilon 12 \log 2\delta$ times.
- 4: return Fraction of times i won

Lean-expanded paper semantic target

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] →
[inst_1 : DecidableEq Arm] →
(epsilon delta : ℝ) →
  FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
  (Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) →
    Bool) →
    ℝ

value:
fun {Arm : Type u_1} [Fintype Arm] [DecidableEq Arm] (epsilon delta : ℝ)
  (pair : FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm)
  (batch :
    Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) → Bool) =>
  FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
  (fun (sampleIndex : ℕ)
    (outcome :
      Fin (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) →
        Bool) =>
      FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
      (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) sampleIndex
      outcome)
  (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm) epsilon delta) batch -
  1 / 2
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Algorithm 6 returns the empirical fraction of wins for one compared pair. This declaration is that finite-batch empirical score minus one half, i.e. the centered estimate used by Algorithm 7; it preserves the source sample budget and introduces no new condition.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput

Paper-local declaration:

FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput

Source locator: cited publication, lines 1182-1199

Verbatim paper-source connection

```
Algorithm 7 S TRONG -T RANSITIVITY-R ANKING
1: inputs
2: Set S, bias  $\epsilon$ , confidence  $\delta$ 
3: for every pair  $\{i, j\}$  such that  $i, j \in S$  do
4:  $\hat{p}_{i,j} \leftarrow \text{E STIMATE -P ROBABILITY}(i, j, \epsilon/2, \delta/n^2)$ 
5:  $\hat{p}_{j,i} \leftarrow 1 - p(i, j)$ 
6: end for
7: ordered set  $T \leftarrow \emptyset$ 
8: while  $|S| > 0$  do
9: if  $\exists e \text{ s.t. } \hat{p}_{e,f} \geq 12 - \epsilon \forall f \in S$  then
10:   Add e at the end of T
11:    $S = S \setminus \{e\}$ 
12: else
13:   Add S at the end of T
14:   return T
15: end if
16: end while
17: return T
```

[Next verbatim source excerpt]

Lemma 20. S TRONG -T RANSITIVITY-R ANKING(S, ϵ, δ) uses $O(n^2 \log n \delta)$ comparisons and w.p. $\geq 1 - \delta$ returns an ϵ -ranking.

Proof. S TRONG -T RANSITIVITY-R ANKING calls E STIMATE -P ROBABILITY for $O(n^2)$ times, once for each pair and each EP $(i, j, \epsilon/2, \delta/n^2)$ uses $O(\epsilon/2 \log n \delta)$ comparisons and hence bound on comparisons follow.

17

[form-feed in source extraction]

w.p. $\geq 1 - \delta/n^2$, E STIMATE -P ROBABILITY($i, j, \epsilon/2, \delta/n^2$) approximates $p_{i,j}$ with $\hat{p}_{i,j}$ such that $|p_{i,j} - \hat{p}_{i,j}| \leq \frac{\epsilon}{2}$

. By the union bound, w.p. $\geq 1 - \delta$, $|p_{i,j} - \hat{p}_{i,j}| \leq 2\epsilon \forall i, j, \in S$. From here we assume that $|p_{i,j} - \hat{p}_{i,j}| \leq 2\epsilon \forall i, j, \in S$ and show that the output is an ϵ -ranking. Let S_t denote the set of remaining elements in S after t elements are removed from S . We first show that for $0 \leq t \leq n - 1$, there is one element e such that $\hat{p}_{e,j} \geq 21 - \epsilon \forall j \in S_t$ and such an element is an ϵ -maximum of S_t . Observe that if there is an element e such that $\hat{p}_{e,j} \geq 21 - \epsilon/2 \forall j \in S_t$ then $p_{e,j} \geq 12 - \epsilon \forall j \in S_t$ and hence e is an ϵ -maximum of S_t . Further recall that for the absolute maximum at^* of S_t , $\text{pat}^*, j \geq 21 \forall j \in S_t$ and hence $\hat{\text{pat}}^*, j \geq 12 - \epsilon/2 \forall j \in S_t$. Therefore there will be at least one element e s.t. $\hat{p}_{e,j} \geq 21 - \epsilon/2$ and such an element will be an ϵ -maximum of S_t . S TRONG -T RANSITIVITY-R ANKING deletes one such element from S_t and adds it to the end of the ordered output set. Since for every t , S TRONG -T RANSITIVITY-R ANKING adds an ϵ -maximum of S_t to the output set, the Lemma follows.

Lean-expanded paper semantic target

```
type:
{Arm : Type u_1} →
[inst : Fintype Arm] → [Nonempty Arm] → [DecidableEq Arm] → (Arm → Arm → ℝ) → ℝ → Fin (Fintype.card Arm) → Arm
```

```
value:
fun {Arm : Type u_1} [Fintype Arm] [Nonempty Arm] [DecidableEq Arm] (estimate : Arm → Arm → ℝ) (epsilon : ℝ)
(slot : Fin (Fintype.card Arm)) =>
FalahatgarEtAl2017MaxingRanking.selectEstimatedThresholdCandidate estimate epsilon
(FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingActiveSet estimate epsilon ↑ slot)
```

Direct retained dependencies

- [FalahatgarEtAl2017MaxingRanking.selectEstimatedThresholdCandidate](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingActiveSet](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The expanded output chooses at each rank the same proof-consistent epsilon/2-qualified arm from the current Algorithm 7 active set, yielding the source ordered removal sequence.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 206-220

Verbatim source input

Lemma 1. For $\epsilon \mu > \epsilon l$, $C\text{OMPARE}(i, j, \epsilon l, \epsilon \mu, \delta)$ uses $\leq (\epsilon \mu - \epsilon l) / (2 \log \delta)$ comparisons and if $\tilde{p}_{i,j} \leq \epsilon l$,

then w.p. $\geq 1 - \delta$, it returns 1, else if $\tilde{p}_{i,j} \geq \epsilon \mu$, w.p. $\geq 1 - \delta$, it returns 2.

Now we present S EQ -E LIMINATE that uses the testing subroutine C OMPARE and finds an ϵ -maximum.

3.1.2 S EQ -E LIMINATE Algorithm

S EQ -E LIMINATE takes a variable set S , selects a random running element $r \in S$ and repeatedly uses C OMPARE($c, r, 0, \epsilon, \delta/n$) to compare r to a random competing element $c \in S \setminus r$. If C OMPARE returns 1 i.e., deems $\tilde{p}_{c,r} \leq 0$, it retains r as the running element and eliminates c from S , but if C OMPARE returns 2 i.e., deems $\tilde{p}_{c,r} \geq \epsilon$, it eliminates r from S and updates c as the new running element.

Expanded PaperInterface specification

```
( $\forall \{ \Omega : \text{Type } u\_1 \}$  (observation :  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) (lower upper delta :  $\mathbb{R}$ ) (outcome :  $\Omega$ ),
  lower < upper  $\rightarrow$ 
    0 < delta  $\rightarrow$ 
      delta  $\leq 1 \rightarrow$ 
        FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime observation
          (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta outcome  $\leq$ 
          FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta  $\wedge$ 
          2 / (upper - lower)  $\wedge$  2 * Real.log (2 / delta)  $\leq$ 
           $\uparrow$  (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)  $\wedge$ 
           $\uparrow$  (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) <
          2 / (upper - lower)  $\wedge$  2 * Real.log (2 / delta) + 1)  $\wedge$ 
( $\forall \{ \Omega : \text{Type } u\_2 \}$  [inst : MeasurableSpace  $\Omega$ ] (law : MeasureTheory.Measure  $\Omega$ ) [MeasureTheory.IsProbabilityMeasure law]
  (observation :  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) (lower upper trueGap delta :  $\mathbb{R}$ ),
  ProbabilityTheory.IndepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) =>
      observation  $\uparrow$  index)
  law  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
  Measurable (observation index))  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
   $\forall^m$  (outcome :  $\Omega$ )  $\partial$ law, observation index outcome  $\in$  Set.lcc 0 1)  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
   $\int$  (x :  $\Omega$ ), observation index x  $\partial$ law = 1 / 2 + trueGap)  $\rightarrow$ 
  trueGap  $\leq$  lower  $\rightarrow$ 
    lower < upper  $\rightarrow$ 
      0 < delta  $\rightarrow$ 
        delta  $\leq 1 \rightarrow$ 
          law.real
            {outcome :  $\Omega$  |
              FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
                (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
                outcome  $\neq$ 
                FalahatgarEtAl2017MaxingRanking.CompareDecision.lower}  $\leq$ 
              delta)  $\wedge$ 
( $\forall \{ \Omega : \text{Type } u\_3 \}$  [inst : MeasurableSpace  $\Omega$ ] (law : MeasureTheory.Measure  $\Omega$ ) [MeasureTheory.IsProbabilityMeasure law]
  (observation :  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) (lower upper trueGap delta :  $\mathbb{R}$ ),
  ProbabilityTheory.IndepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) =>
      observation  $\uparrow$  index)
  law  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
  Measurable (observation index))  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
   $\forall^m$  (outcome :  $\Omega$ )  $\partial$ law, observation index outcome  $\in$  Set.lcc 0 1)  $\rightarrow$ 
( $\forall$  index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
   $\int$  (x :  $\Omega$ ), observation index x  $\partial$ law = 1 / 2 + trueGap)  $\rightarrow$ 
  upper  $\leq$  trueGap  $\rightarrow$ 
    lower < upper  $\rightarrow$ 
      0 < delta  $\rightarrow$ 
        delta  $\leq 1 \rightarrow$ 
          law.real
            {outcome :  $\Omega$  |
              FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
                (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
                outcome  $\neq$ 
                FalahatgarEtAl2017MaxingRanking.CompareDecision.upper}  $\leq$ 
              delta)
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)

- [FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {Ω : Type u_1} (observation : ℕ → Ω → ℝ) (lower upper delta : ℝ) (outcome : Ω),
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime observation
    (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta outcome ≤
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta ∧
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) ≤
  ↑(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) ∧
  ↑(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) <
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) + 1) ∧
(∀ {Ω : Type u_2} [inst : MeasurableSpace Ω] (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law]
  (observation : ℕ → Ω → ℝ) (lower upper trueGap delta : ℝ),
  ProbabilityTheory.IndepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) =>
      observation ↑index)
  law →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    Measurable (observation index)) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.lcc 0 1) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap) →
  trueGap ≤ lower →
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
        outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.lower} ≤
    delta) ∧
  ∀ {Ω : Type u_3} [inst : MeasurableSpace Ω] (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law]
    (observation : ℕ → Ω → ℝ) (lower upper trueGap delta : ℝ),
  ProbabilityTheory.IndepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) =>
      observation ↑index)
  law →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    Measurable (observation index)) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.lcc 0 1) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap) →
  upper ≤ trueGap →
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
        outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.upper} ≤
    delta)

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma1_compare_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The single expanded Spec contains all three Main Lemma 1 clauses: the ceiling-corrected Compare cap, failure at most delta for the lower decision when the true gap is at most the lower threshold, and failure at most delta for the upper decision when it is at least the upper threshold.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 239-247

Verbatim source input

Theorem 2. S EQ -E LIMINATE(S , ϵ , δ) uses $O\left(\frac{\log |S|}{\epsilon^2}\right)$ comparisons, and w.p. $\geq 1 - \delta$ outputs an ϵ -maximum.

4

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
(∀ (arm : Arm), preferenceGap arm arm = 0) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
0 < epsilon →
0 < delta →
delta ≤ 1 →
1 - delta ≤
FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateEpsilonMaximumProbability
preferenceGap hprobability epsilon delta) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope armCount epsilon delta ≤
3 * ↑armCount * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateEpsilonMaximumProbability](#)
- [FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(epsilon delta : ℝ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
(∀ (arm : Arm), preferenceGap arm arm = 0) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
0 < epsilon →
0 < delta →
delta ≤ 1 →
1 - delta ≤
FalahatgarEtAl2017MaxingRanking.canonicalFreshUniformSeqEliminateEpsilonMaximumProbability
preferenceGap hprobability epsilon delta) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
FalahatgarEtAl2017MaxingRanking.seqEliminateSourceComparisonEnvelope armCount epsilon delta ≤
3 * ↑armCount * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem2_seqEliminate_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target states Main Theorem 2 on the source finite SST comparison model: the canonical fresh Seq-Eliminate output is an epsilon-maximum with probability at least $1-\delta$, and its complete source envelope has the stated $O(n \epsilon^{-2} \log(n/\delta))$ form.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Review row 3

Source locator: cited publication, lines 279-281

Verbatim source input

Lemma 3. P ICK -A NCHOR(S, n', ϵ, δ) uses $O(nn' \epsilon^2 \log 16 \log nn' \delta)$ comparisons and w.p. $\geq 1 - \delta$, outputs an (ϵ, n') -good anchor element.

2

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (cutoff : ℕ) (delta epsilon : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hcutoff : 0 < cutoff),
cutoff ≤ Fintype.card Arm →
∀ (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1),
0 < epsilon →
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
(∀ (arm : Arm), preferenceGap arm arm = 0) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
1 - delta ≤
FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability cutoff delta epsilon
(fun
(sample :
AppliedModelingLib.finiteFreshList Arm
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)
∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw sample
(FalahatgarEtAl2017MaxingRanking.pickAnchorSourceSampleCount_pos cutoff delta hcutoff hdelta
hdeltaLeOne)
preferenceGap hprobability epsilon
(delta / 2 /
↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)))
(fun
(sample :
AppliedModelingLib.finiteFreshList Arm
(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)
∅) =>
FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation sample epsilon
(delta / 2 /
↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)))
preferenceGap hcutoff hdelta hdeltaLeOne) ∧
∀ {Arm : Type u_2} [inst : Fintype Arm] [Nonempty Arm] (cutoff : ℕ) (epsilon delta : ℝ),
0 < cutoff →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑ ((FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta - 1) *
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon
(delta / 2 /
↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta))) ≤
5 * (↑ (Fintype.card Arm) / ↑ cutoff * Real.log (2 / delta) + 1) *
(1 + Real.log ((↑ (Fintype.card Arm) / ↑ cutoff * Real.log (2 / delta) + 1) / delta)) /
epsilon ^ 2
```

Retained governing declarations

- [AppliedModelingLib.finiteFreshList](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (cutoff : ℕ) (delta epsilon : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hcutoff : 0 < cutoff),
cutoff ≤ Fintype.card Arm →
∀ (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1),
```

```

0 < epsilon →
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
(∀ (arm : Arm), preferenceGap arm arm = 0) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
1 - delta ≤
  FalahatgarEtAl2017MaxingRanking.freshPickAnchorUniformGoodAnchorProbability cutoff delta epsilon
  (fun
    (sample :
      AppliedModelingLib.finiteFreshList Arm
      (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)
      () =>
        FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorOutcomeLaw sample
        (FalahatgarEtAl2017MaxingRanking.pickAnchorSourceSampleCount_pos cutoff delta hcutoff hdelta
         hdeltaLeOne)
        preferenceGap hprobability epsilon
        (delta / 2 /
         ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)))
    (fun
      (sample :
        AppliedModelingLib.finiteFreshList Arm
        (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)
        () =>
          FalahatgarEtAl2017MaxingRanking.canonicalFreshPickAnchorObservation sample epsilon
          (delta / 2 /
           ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta)))
      preferenceGap hcutoff hdelta hdeltaLeOne) ∧
  preferenceGap hcutoff hdelta hdeltaLeOne) ∧
∀ {Arm : Type u_2} [inst : Fintype Arm] [Nonempty Arm] (cutoff : ℕ) (epsilon delta : ℝ),
0 < cutoff →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑ ((FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta - 1) *
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon
  (delta / 2 /
   ↑ (FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta))) ≤
5 * (↑ (Fintype.card Arm) / ↑ cutoff * Real.log (2 / delta) + 1) *
(1 + Real.log ((↑ (Fintype.card Arm) / ↑ cutoff * Real.log (2 / delta) + 1) / delta)) /
epsilon ^ 2

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma3_pickAnchor_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target preserves Pick-Anchor's uniform sample without replacement, Seq-Eliminate source allocation, SST good-anchor conclusion with probability at least 1-delta, and the stated source comparison-rate envelope. Its finite sample cap and ceilings make the algorithm's displayed rate precise rather than changing its scope.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 282-286

Verbatim source input

Remark 4. Note that P ICK -A NCHOR(S, cn, ϵ, δ) uses $O_c(\epsilon^{12}(\log 16))$ comparisons where the constant depends only on c but not on n . Hence it is advantageous to use this method to pick near-maximum element when n is large.

We now present P RUNE that takes an anchor element as input and prunes the set S using the anchor.

Expanded PaperInterface specification

```
∀ {Arm : Type u_1} [inst : Fintype Arm] [Nonempty Arm] (cutoff : ℕ) (c epsilon delta : ℝ),
  0 < cutoff →
  0 < epsilon →
  epsilon ≤ 1 →
  0 < c →
  0 < delta →
  delta ≤ 1 →
  ↑(Fintype.card Arm) / ↑cutoff ≤ 1 / c →
  ↑((FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta - 1) *
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon
    (delta / 2 /
      ↑(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta))) ≤
  5 * (1 / c * Real.log (2 / delta) + 1) * (1 + Real.log ((1 / c * Real.log (2 / delta) + 1) / delta)) /
  epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount](#)

Lean-elaborated claim roles

1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. parameter: ℕ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: ℝ
8. assumption: 0 < cutoff
9. assumption: 0 < epsilon
10. assumption: epsilon ≤ 1
11. assumption: 0 < c
12. assumption: 0 < delta
13. assumption: delta ≤ 1
14. assumption: ↑(Fintype.card Arm) / ↑cutoff ≤ 1 / c
15. conclusion: ↑((FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta - 1) *
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget 0 epsilon
(delta / 2 / ↑(FalahatgarEtAl2017MaxingRanking.pickAnchorSampleCount (Fintype.card Arm) cutoff delta))) ≤
5 * (1 / c * Real.log (2 / delta) + 1) * (1 + Real.log ((1 / c * Real.log (2 / delta) + 1) / delta)) / epsilon ^ 2

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.remark4_pickAnchorLinearCutoff_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded target specializes the Pick-Anchor resource theorem to $n/n' \leq 1/c$ and proves a ceiling-corrected bound depending on c , ϵ , and δ but not on n , exactly the quantitative content of Remark 4.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 330-336

Verbatim source input

Lemma 5. If $n' \geq 6n \log n$, $\delta \geq n1$ and a is an $(\epsilon1, n')$ -good anchor element, then w.p. $\geq 1 - 2\delta$,

$P \text{ RUNE}(S, a, n', \epsilon1, \epsilon u, \delta)$ uses $O(\frac{1}{2 \log \delta} (\epsilon u - \epsilon1))$ comparisons and outputs a set of size less than

$2n'$. Further if a is not an ϵu -maximum of S then w.p. $\geq 1 - 2\delta$, the output set contains an absolute maximum element of S .

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For the finite centered SST Prune model, under the displayed weak cutoff $\sqrt{6n \log n} \leq n'$, confidence $1/n \leq \delta$ with $0 < \delta \leq 1/2$, threshold $\hookrightarrow$ separation, and good-anchor premises, the source-scheduled stopped execution has probability at least $1-\delta/2$ of returning at most $2n'$ arms while $\hookrightarrow$ using no more than $16n(2(\epsilon u - \epsilon1)^2 + 1)(1 + \log(1/\delta))$ comparisons. If the anchor is not an ϵu -maximum, the same $\hookrightarrow$ source-scheduled Prune execution retains an internally chosen absolute maximum with probability at least $1-\delta/2$.

Archival source anchor: cited publication, lines 330-336

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} [inst : Fintype Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (anchor : Arm)
(lower upper delta : ℝ) (cutoff : ℕ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
0 ≤ upper →
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
lower < upper →
0 < delta →
delta ≤ 1 / 2 →
√(6 * ↑(Fintype.card Arm) * Real.log ↑(Fintype.card Arm)) ≤ ↑cutoff →
1 / ↑(Fintype.card Arm) ≤ delta →
let roundCount : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm);
let maxBatch : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch lower upper delta;
have hbudget :
  ∀ round < roundCount,
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
    (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
      maxBatch :=
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget_le_sourcePruneMaxBatch_of_source_round lower
upper delta;
have outcomeLaw :
AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch) :=
FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
roundCount anchor lower upper delta maxBatch hbudget;
have decision :
  ℕ →
  Finset Arm →
  FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch →
  Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision :=
FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision roundCount lower upper delta
maxBatch hbudget;
(1 - delta / 2 ≤
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceLaw cutoff Finset.univ outcomeLaw
decision roundCount)
fun (state : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount) =>
state.1.card ≤ 2 * cutoff ∧
↑(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount roundCount cutoff lower
upper delta state.2) ≤
FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound (Fintype.card Arm)
lower upper delta) ∧
(→FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap upper anchor →
1 - delta / 2 ≤
AppliedModelingLib.PreferenceRL.pmfProbClassical
(FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw Finset.univ outcomeLaw cutoff
decision roundCount)
fun (active : Finset Arm) =>
∃ (maximum : Arm),
FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum ∧
maximum ∈ active)

```

Retained governing declarations

- [AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel](#)
- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [AppliedModelingLib.pmfProb](#)

- [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Arm
3. conclusion: ∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (anchor : Arm)
(lower upper delta : ℝ) (cutoff : ℕ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
0 ≤ upper →
  FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
  lower < upper →
  0 < delta →
  delta ≤ 1 / 2 →
  √(6 * ↑(Fintype.card Arm) * Real.log ↑(Fintype.card Arm)) ≤ ↑cutoff →
  1 / ↑(Fintype.card Arm) ≤ delta →
  let roundCount : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm);
  let maxBatch : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch lower upper delta;
  have hbudget :
    ∀ round < roundCount,
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
        (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
        maxBatch :=
    FalahatgarEtAl2017MaxingRanking.fixedSampleBudget_le_sourcePruneMaxBatch_of_source_round lower
      upper delta;
  have outcomeLaw :
    AppliedModelingLib.PreferenceRL.AdaptiveOutcomeKernel (Finset Arm)
      (FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch) :=
    FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
      roundCount anchor lower upper delta maxBatch hbudget;
  have decision :
    ℕ →
    Finset Arm →
    FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome Arm maxBatch →
    Arm → FalahatgarEtAl2017MaxingRanking.CompareDecision :=
    FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision roundCount lower upper delta maxBatch
      hbudget;
(1 - delta / 2 ≤
  AppliedModelingLib.pmfProb
    (FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceLaw cutoff Finset.univ outcomeLaw
      decision roundCount)
  fun (state : FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm roundCount) =>
    state.1.card ≤ 2 * cutoff ∧
    ↑(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount roundCount cutoff lower
      upper delta state.2) ≤
    FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound (Fintype.card Arm)
      lower upper delta) ∧
  (¬FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap upper anchor →
    1 - delta / 2 ≤
    AppliedModelingLib.PreferenceRL.pmfProbClassical
      (FalahatgarEtAl2017MaxingRanking.freshStoppedPruneActiveLaw Finset.univ outcomeLaw cutoff
        decision roundCount)
    fun (active : Finset Arm) =>
      ∃ (maximum : Arm),
        FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum ∧ maximum ∈ active)

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma5_prune_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: The checked statement retains Lemma 5's cutoff, confidence, rate, and conditional maximum-retention conclusions. Its output endpoint is at most $2n'$ because Algorithm 2 can stop at equality.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 363-364

Verbatim source input

Theorem 6. $W.p. \geq 1 - \delta$, O PT-M AXIMIZE(S, ϵ, δ) uses $O(\epsilon n^2 \log 1/\delta)$ comparisons and outputs an ϵ -maximum.

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (epsilon delta : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (hdelta : 0 < delta)
(hdeltaLeOne : delta ≤ 1)
(hantisymmetric : ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second),
(∀ (arm : Arm), preferenceGap arm arm = 0) →
  ∀ (hsst : FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap),
  0 < epsilon →
  have x : DecidableEq Arm := Classical.decEq Arm;
  have maximum : Arm :=
    Classical.choose
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_1 preferenceGap hantisymmetric hsst);
  1 - delta ≤
    FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability
      (epsilon / 3) (2 * epsilon / 3) epsilon delta
      (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch (epsilon / 3) (2 * epsilon / 3)
        delta)
      preferenceGap maximum hprobability hdelta hdeltaLeOne
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_3 epsilon delta)) ∧
  ∀ {epsilon delta : ℝ},
  0 < epsilon →
  0 < delta →
  delta ≤ 1 →
  (fun (armCount : ℕ) =>
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3TotalRateEnvelope armCount epsilon
      delta) =O[Filter.atTop]
    fun (armCount : ℕ) => ↑armCount * (1 + Real.log (1 / delta)) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3TotalRateEnvelope](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (epsilon delta : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (hdelta : 0 < delta)
(hdeltaLeOne : delta ≤ 1)
(hantisymmetric : ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second),
(∀ (arm : Arm), preferenceGap arm arm = 0) →
  ∀ (hsst : FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap),
  0 < epsilon →
  have x : DecidableEq Arm := Classical.decEq Arm;
  have maximum : Arm :=
    Classical.choose
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_1 preferenceGap hantisymmetric hsst);
  1 - delta ≤
    FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAlgorithm3SourcePopulationJointEnvelopeProbability
      (epsilon / 3) (2 * epsilon / 3) epsilon delta
      (FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch (epsilon / 3) (2 * epsilon / 3)
        delta)
      preferenceGap maximum hprobability hdelta hdeltaLeOne
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_3 epsilon delta)) ∧
  ∀ {epsilon delta : ℝ},
  0 < epsilon →
  0 < delta →
  delta ≤ 1 →
  (fun (armCount : ℕ) =>
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3TotalRateEnvelope armCount epsilon
      delta) =O[Filter.atTop]
    fun (armCount : ℕ) => ↑armCount * (1 + Real.log (1 / delta)) / epsilon ^ 2
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem6_optMaximize_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The checked theorem combines the source Seq-Eliminate fallback and the Algorithm 3 branch, preserving the probability-one-minus-delta epsilon-maximum conclusion and the stated asymptotic comparison rate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Review row 7

Source locator: cited publication, lines 408-414

Verbatim source input

Theorem 7. There exists a model that satisfies SST for which any algorithm requires $\Omega(n^2)$ comparisons to find a $1/4$ -ranking with probability $\geq 7/8$.

We also present a trivial ε -ranking algorithm in Appendix B.2 that for any stochastic model with ranking (Weak Stochastic Transitivity), uses $O(n^2 \log n)$ comparisons and outputs an ε -ranking w.p. $\geq 1 - \delta$.

Expanded PaperInterface specification

```

∀ {Seed : Type u_1} [inst : Fintype Seed] [Nonempty Seed],
have x : DecidableEq Seed := Classical.decEq Seed;
∀ {n comparisonBudget : ℕ} {mu : ℝ} (seedLaw : PMF Seed) (hmu : 0 < mu) (hmuHalf : mu ≤ 1 / 2) (hcard : 2 ≤ n),
mu ≤ 1 / ↑n ^ 10 →
↑comparisonBudget ≤ ↑n ^ 2 / 20 →
  ∀ (procedure : Seed → FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget),
  ∃ (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool),
    ¬ 7 / 8 ≤
      FalahatgarEtAl2017MaxingRanking.theorem7SeededSmallMuInstanceSuccessProbability seedLaw mu
      (FalahatgarEtAl2017MaxingRanking.theorem7_rankingLowerBoundSpec._proof_4 hmu) hmuHalf hcard procedure
      coordinate orientation

```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure](#)
- [FalahatgarEtAl2017MaxingRanking.theorem7SeededSmallMuInstanceSuccessProbability](#)
- [FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Seed
3. assumption: Nonempty Seed
4. conclusion: ∀ {n comparisonBudget : ℕ} {mu : ℝ} (seedLaw : PMF Seed) (hmu : 0 < mu) (hmuHalf : mu ≤ 1 / 2) (hcard : 2 ≤ n),
mu ≤ 1 / ↑n ^ 10 →
  ↑comparisonBudget ≤ ↑n ^ 2 / 20 →
    ∀ (procedure : Seed → FalahatgarEtAl2017MaxingRanking.Theorem7AdaptiveRankingProcedure n comparisonBudget),
    ∃ (coordinate : FalahatgarEtAl2017MaxingRanking.unorderedComparisonCoordinate (Fin n)) (orientation : Bool),
      ¬ 7 / 8 ≤
        FalahatgarEtAl2017MaxingRanking.theorem7SeededSmallMuInstanceSuccessProbability seedLaw mu
        (FalahatgarEtAl2017MaxingRanking.theorem7_rankingLowerBoundSpec._proof_4 hmu) hmuHalf hcard procedure
        coordinate orientation

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem7_rankingLowerBound_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Supplement B.1 makes the exceptional pair a random relabeling and thereby proves the standard worst-case lower-bound order retained here: every finite-random-tape procedure below $n^2/20$ fails on some relabeled SST hard instance with success below $7/8$. This is the documented clarification of the short display, not a theorem weakening.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 431-448

Verbatim source input

Theorem 8. There exists an algorithm that uses $O(\epsilon^2 \log 1/\delta)$ comparisons and w.p. $\geq 1 - \delta$, outputs an ϵ -Borda maximum.

7

[form-feed in source extraction]

For ϵ -Borda ranking, we note that if we compare an element e with $\epsilon^2 \log 2n$

δ

random elements, w.p.

$\geq 1 - \delta/n$, the fraction of times e wins approximates the Borda score of e to an additive error of ϵ

. Ranking based on these approximate scores results in an ϵ -Borda ranking. We present BORDA-RANKING in C.1 that uses $2n$

ϵ^2

$\log 2n$

δ

comparisons and w.p. $\geq 1 - \delta$ outputs an ϵ -Borda ranking. Proof in C.1.

Expanded PaperInterface specification

```
∃ (coefficient : ℝ),
  ∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (epsilon delta : ℝ),
    0 < epsilon →
      epsilon ≤ 1 →
        0 < delta →
          delta ≤ 1 →
            ∃ (algorithm : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm),
              ∀ (winProbability : Arm → Arm → ℝ)
                (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability),
                1 - delta ≤
                  FalahatgarEtAl2017MaxingRanking.finiteAdaptiveBordaMaxingSuccessProbability winProbability
                    hprobability epsilon algorithm ∧
                  ∀ (rewards : Fin algorithm.pullBudget → Bool),
                    ↑ (algorithm.executionComparisonCount rewards) ≤
                      coefficient * ↑ (Fintype.card Arm) * (1 + Real.log (1 / (delta / 2))) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm](#)
- [FalahatgarEtAl2017MaxingRanking.finiteAdaptiveBordaMaxingSuccessProbability](#)

Lean-elaborated claim roles

```
1. conclusion: ∃ (coefficient : ℝ),
  ∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (epsilon delta : ℝ),
    0 < epsilon →
      epsilon ≤ 1 →
        0 < delta →
          delta ≤ 1 →
            ∃ (algorithm : FalahatgarEtAl2017MaxingRanking.FiniteAdaptiveBordaMaxingAlgorithm Arm),
              ∀ (winProbability : Arm → Arm → ℝ)
                (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability),
                1 - delta ≤
                  FalahatgarEtAl2017MaxingRanking.finiteAdaptiveBordaMaxingSuccessProbability winProbability
                    hprobability epsilon algorithm ∧
                  ∀ (rewards : Fin algorithm.pullBudget → Bool),
                    ↑ (algorithm.executionComparisonCount rewards) ≤
                      coefficient * ↑ (Fintype.card Arm) * (1 + Real.log (1 / (delta / 2))) / epsilon ^ 2
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem8_bordaMaxing_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target has the source Theorem 8 quantifier order: one finite adaptive Borda-maxing algorithm is selected before every admissible win matrix, succeeds with probability at least $1 - \delta$, and has $O(n \epsilon^2 \log(1/\delta))$ realized comparison cost. The finite policy and uniform-opponent bridge make the cited bandit reduction explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 449-453

Verbatim source input

Theorem 9. B ORDA -R ANKING(S, ϵ, δ) uses $2n \cdot \frac{\log 2n}{\epsilon^2}$ comparisons and w.p. $\geq 1 - \delta$ outputs an ϵ -Borda ranking.

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
have x : DecidableEq Arm := Classical.decEq Arm;
∃ (rankingRule : (Arm → ℝ) → Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule rankingRule ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon
delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) →
Bool |
¬FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking winProbability epsilon
(rankingRule fun (arm : Arm) =>
FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) arm)
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta)
labelTable)} ≤
delta) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑(FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount armCount epsilon delta) ≤
3 * ↑armCount * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking](#)
- [FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule](#)
- [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
- [FalahatgarEtAl2017MaxingRanking.bordaSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
have x : DecidableEq Arm := Classical.decEq Arm;
∃ (rankingRule : (Arm → ℝ) → Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule rankingRule ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon
delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) →
Bool |
¬FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking winProbability epsilon
(rankingRule fun (arm : Arm) =>
FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
```

```

(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) arm)
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta)
labelTable)} ≤
delta) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑ (FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount armCount epsilon delta) ≤
3 * ↑ armCount * (1 + Real.log (↑ armCount / delta)) / epsilon ^ 2

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem9_bordaRanking_proof

Recorded source-to-Spec screening Verdict: matches
Reason: The target preserves Main Theorem 9’s Algorithm-9 epsilon-Borda-ranking success event at probability 1-delta and its displayed $2n \epsilon^{-2} \log(2n/\delta)$ count, with the source finite sampling schedule and an explicit empirical tie-safe ranking rule.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 10

Source locator: cited publication, lines 764-768

Verbatim source input

Lemma 10. For $\epsilon_u > \epsilon_l \geq 0$, $\text{COMPARE}(i, j, \epsilon_l, \epsilon_u, \delta)$ uses $\leq (\epsilon_u - \epsilon_l)^2 \log 26$ comparisons.

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} (observation : ℕ → Ω → ℝ) (lower upper delta : ℝ) (outcome : Ω),
  0 ≤ lower →
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime observation
    (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta outcome ≤
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta ∧
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) ≤
  ↑ (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) ∧
  ↑ (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) <
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) + 1
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: ℕ → Ω → ℝ
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℝ
6. parameter: Ω
7. assumption: 0 ≤ lower
8. assumption: lower < upper
9. assumption: 0 < delta
10. assumption: delta ≤ 1
11. conclusion: FalahatgarEtAl2017MaxingRanking.adaptiveCompareStoppingTime observation
  (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta outcome ≤
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta ∧
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) ≤
  ↑ (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) ∧
  ↑ (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) <
  2 / (upper - lower) ^ 2 * Real.log (2 / delta) + 1
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma10_compareCap_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded target has the source premises $\text{upper} > \text{lower} \geq 0$, caps Compare at the natural ceiling of $2/(\text{upper}-\text{lower})^2 \log(2/\text{delta})$, and records the exact ceiling interval from the displayed real count to that count plus one.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 779

Verbatim source input

Lemma 11. For $\epsilon_u > \epsilon_l \geq 0$, if $\bar{p}_{i,j} \leq \epsilon_l$, then w.p. $\geq 1 - \delta$, $C_{OMPARE}(i, j, \epsilon_l, \epsilon_u, \delta)$ outputs 1.

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law]
  (observation : ℕ → Ω → ℝ) (lower upper trueGap delta : ℝ),
  ProbabilityTheory.indepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) => observation ↑ index)
  law →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta, Measurable (observation index)) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.Icc 0 1) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap) →
  trueGap ≤ lower →
  0 ≤ lower →
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
        outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.lower} ≤
    delta
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: MeasurableSpace Ω
3. parameter: MeasureTheory.Measure Ω
4. assumption: MeasureTheory.IsProbabilityMeasure law
5. parameter: ℕ → Ω → ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. assumption: ProbabilityTheory.indepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) => observation ↑ index) law
11. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta, Measurable (observation index)
12. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.Icc 0 1
13. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap
14. assumption: trueGap ≤ lower
15. assumption: 0 ≤ lower
16. assumption: lower < upper
17. assumption: 0 < delta
18. assumption: delta ≤ 1
19. conclusion: law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
        outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.lower} ≤
    delta
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma11_compareLower_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded target assumes fresh bounded independent comparisons with centered mean at most the lower threshold and bounds the probability that Compare fails to output the source lower decision by delta.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 811

Verbatim source input

Lemma 12. For $\epsilon_u > \epsilon_l \geq 0$, if $\bar{p}_{i,j} \geq \epsilon_u$, then w.p. $\geq 1 - \delta$, $C_{OMPARE}(i, j, \epsilon_l, \epsilon_u, \delta)$ outputs 2.

Expanded PaperInterface specification

```

∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law]
  (observation : ℕ → Ω → ℝ) (lower upper trueGap delta : ℝ),
  ProbabilityTheory.indepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) => observation ↑ index)
    law →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta, Measurable (observation index)) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.Icc 0 1) →
  (∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap) →
  upper ≤ trueGap →
  0 ≤ lower →
  lower < upper →
  0 < delta →
  delta ≤ 1 →
  law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta
        outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.upper} ≤
    delta

```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptiveCompare](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: MeasurableSpace Ω
3. parameter: MeasureTheory.Measure Ω
4. assumption: MeasureTheory.IsProbabilityMeasure law
5. parameter: ℕ → Ω → ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. assumption: ProbabilityTheory.indepFun
    (fun (index : Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta)) => observation ↑ index) law
11. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta, Measurable (observation index)
12. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∀m (outcome : Ω) ∂law, observation index outcome ∈ Set.Icc 0 1
13. assumption: ∀ index < FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta,
    ∫ (x : Ω), observation index x ∂law = 1 / 2 + trueGap
14. assumption: upper ≤ trueGap
15. assumption: 0 ≤ lower
16. assumption: lower < upper
17. assumption: 0 < delta
18. assumption: delta ≤ 1
19. conclusion: law.real
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptiveCompare observation
        (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper delta) lower upper delta outcome ≠
        FalahatgarEtAl2017MaxingRanking.CompareDecision.upper} ≤
    delta

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma12_compareUpper_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The expanded target assumes fresh bounded independent comparisons with centered mean at least the upper threshold and bounds the probability that Compare fails to output the source upper decision by delta.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 923

Verbatim source input

Lemma 13. If $\bar{p}_e, a \geq \varepsilon u$, then w.p. $\geq 1 - \delta/2$, the output set of $P_{RUNE}(S, a, n', \varepsilon l, \varepsilon u, \delta)$ contains e .

Expanded PaperInterface specification

```

∀ {Ω : Type u_1} {Arm : Type u_2} [inst : MeasurableSpace Ω],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law] (roundCount : ℕ)
  (observation : ℕ → Arm → ℕ → Ω → ℝ) (active : Finset Arm) (kept : Arm) (lower upper centeredGap delta : ℝ),
kept ∈ active →
  (∀ round < roundCount,
    ProbabilityTheory.indepFun
      (fun
        (sample :
          Fin
            (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
              (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round))) =>
          observation round kept ↑sample)
      law) →
  (∀ round < roundCount,
    ∀
      sample <
        FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
      Measurable (observation round kept sample)) →
  (∀ round < roundCount,
    ∀
      sample <
        FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
      ∀m (outcome : Ω) ∂law, observation round kept sample outcome ∈ Set.lcc 0 1) →
  (∀ round < roundCount,
    ∀
      sample <
        FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
          (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
      f (x : Ω), observation round kept sample x ∂law = 1 / 2 + centeredGap) →
  (∀ round < roundCount,
    MeasurableSet
      {outcome : Ω |
        FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision observation lower upper delta round kept
          outcome =
            FalahatgarEtAl2017MaxingRanking.CompareDecision.upper}) →
    upper ≤ centeredGap →
    lower < upper →
    0 < delta →
    delta ≤ 1 →
    1 - delta / 2 ≤
      law.real
        {outcome : Ω |
          kept ∈
            FalahatgarEtAl2017MaxingRanking.indexedPruneRounds roundCount
              (FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision observation lower upper delta)
              outcome active})

```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CompareDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.indexedPruneRounds](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: MeasurableSpace Ω
4. conclusion: ∀ (law : MeasureTheory.Measure Ω) [MeasureTheory.IsProbabilityMeasure law] (roundCount : ℕ)
  (observation : ℕ → Arm → ℕ → Ω → ℝ) (active : Finset Arm) (kept : Arm) (lower upper centeredGap delta : ℝ),
kept ∈ active →
  (∀ round < roundCount,
    ProbabilityTheory.indepFun
      (fun
        (sample :
          Fin
            (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
              (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round))) =>
          observation round kept ↑sample)
      law) →

```

```

    observation round kept ↑ sample)
law) →
(∀ round < roundCount,
  ∀
    sample <
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
    Measurable (observation round kept sample)) →
(∀ round < roundCount,
  ∀
    sample <
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
    ∀m (outcome : Ω) ∂law, observation round kept sample outcome ∈ Set.Icc 0 1) →
(∀ round < roundCount,
  ∀
    sample <
      FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
      (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round),
    ∫ (x : Ω), observation round kept sample x ∂law = 1 / 2 + centeredGap) →
(∀ round < roundCount,
  MeasurableSet
    {outcome : Ω |
      FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision observation lower upper delta round kept
      outcome =
      FalahatgarEtAl2017MaxingRanking.CompareDecision.upper}) →
upper ≤ centeredGap →
lower < upper →
0 < delta →
delta ≤ 1 →
1 - delta / 2 ≤
law.real
{outcome : Ω |
  kept ∈
    FalahatgarEtAl2017MaxingRanking.indexedPruneRounds roundCount
    (FalahatgarEtAl2017MaxingRanking.adaptivePruneDecision observation lower upper delta)
    outcome active}

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma13_pruneRetention_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The source’s protected-arm retention conclusion is retained: under the upper-gap hypothesis and its geometric delta schedule, the finite Prune execution keeps the arm with probability at least 1-delta/2. The target makes the source iid observation model and finite rounds explicit without changing that event or endpoint.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 943-948

Verbatim source input

Lemma 14. If $n' \geq 8 \log 2 n$, $n_1 \leq \delta \leq nn$ and a is an $(\epsilon l, n')$ -good anchor element then w.p. $\geq 1 - \delta/2$, $P \text{ RUNE}(S, a, n', \epsilon l, \epsilon u, \delta)$ uses $O(\epsilon u - \epsilon l \log 16)$ comparisons and outputs a set of size at most $2n'$.

Expanded PaperInterface specification

```
∀ {Arm : Type u_1} [inst : Fintype Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (active : Finset Arm)
(anchor : Arm) (lower upper delta : ℝ) (cutoff : ℕ),
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
lower < upper →
0 < delta →
delta ≤ 1 →
8 * Real.log ↑(Fintype.card Arm) ^ 2 ≤ ↑cutoff →
1 / ↑(Fintype.card Arm) ≤ delta →
delta ≤ ↑cutoff / ↑(Fintype.card Arm) →
(FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw preferenceGap hprobability active anchor
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
(delta / 4))).toMeasure.real
{batchTable :
↑active →
Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper (delta / 4)) → Bool |
¬(FalahatgarEtAl2017MaxingRanking.pruneRound active
(FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision active lower upper
(delta / 4) batchTable)).card ≤
2 * cutoff} ≤
delta / 2 ∧
↑(active.card * FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper (delta / 4)) ≤
↑active.card * (2 / (upper - lower) ^ 2 * Real.log (2 / (delta / 4)) + 1)
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.pruneRound](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Fintype Arm
3. conclusion: ∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (active : Finset Arm)
(anchor : Arm) (lower upper delta : ℝ) (cutoff : ℕ),
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
lower < upper →
0 < delta →
delta ≤ 1 →
8 * Real.log ↑(Fintype.card Arm) ^ 2 ≤ ↑cutoff →
1 / ↑(Fintype.card Arm) ≤ delta →
delta ≤ ↑cutoff / ↑(Fintype.card Arm) →
(FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundBatchLaw preferenceGap hprobability active anchor
(FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper (delta / 4))).toMeasure.real
{batchTable :
↑active →
Fin (FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper (delta / 4)) → Bool |
¬(FalahatgarEtAl2017MaxingRanking.pruneRound active
(FalahatgarEtAl2017MaxingRanking.canonicalPruneRoundDecision active lower upper
(delta / 4) batchTable)).card ≤
2 * cutoff} ≤
delta / 2 ∧
↑(active.card * FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper (delta / 4)) ≤
↑active.card * (2 / (upper - lower) ^ 2 * Real.log (2 / (delta / 4)) + 1)
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma14_pruneOneRound_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target preserves Supplement Lemma 14's good-anchor, cutoff, and confidence domain, its first-round at-most- $2n'$ conclusion with failure at most $\delta/2$, and the displayed finite Compare budget. The canonical finite Bernoulli batch law is the source comparison model made explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 974-984

Verbatim source input

Lemma 15. If $n' > 6n \log n$, $\delta \geq nn$ and a is an $(\epsilon l, n')$ -good anchor element, then w.p. $\geq 1 - 2\delta$, $\text{PRUNE}(S, a, n', \epsilon l, \epsilon u, \delta)$ uses $O(\epsilon u - \epsilon n^2 \log 16)$ comparisons and outputs a set of size less than $2n'$.

14

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For the finite centered SST Prune model, under the displayed strict cutoff $\sqrt{6n \log n} < n'$, confidence $n'/n \leq \delta$ with $0 < \delta \leq 1/2$, threshold $\hookrightarrow$ separation, and good-anchor premises, the source-scheduled stopped execution has probability at least $1 - \delta/2$ of returning at most $2n'$ arms within $\hookrightarrow$ the checked Prune comparison-rate bound.

Archival source anchor: cited publication, lines 974-984

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} [inst : Fintype Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (anchor : Arm)
(lower upper delta : ℝ) (cutoff : ℕ) (initial : Finset Arm),
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
lower < upper →
0 < delta →
delta ≤ 1 / 2 →
√(6 * ↑(Fintype.card Arm) * Real.log ↑(Fintype.card Arm)) < ↑cutoff →
↑cutoff / ↑(Fintype.card Arm) ≤ delta →
let maxBatch : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch lower upper delta;
have hbudget :
  ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
  maxBatch :=
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget_le_sourcePruneMaxBatch_of_source_round lower upper
delta;
1 - delta / 2 ≤
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw preferenceGap lower
(FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor (Fintype.card Arm) cutoff
delta)
cutoff anchor initial
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) anchor lower upper
delta maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) lower upper delta
maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)))
fun
(stateFailure :
  FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm
  (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) ×
  Bool) =>
stateFailure.1.1.card ≤ 2 * cutoff ∧
↑(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) cutoff lower
upper delta stateFailure.1.2) ≤
FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound (Fintype.card Arm) lower
upper delta
```

Retained governing declarations

- [AppliedModelingLib.pmfProb](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.GoodAnchor](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw](#)

- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcome](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneTaggedOutcomeFintype](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound](#)
- [FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Arm
3. conclusion: ∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (anchor : Arm)
(lower upper delta : ℝ) (cutoff : ℕ) (initial : Finset Arm),
FalahatgarEtAl2017MaxingRanking.GoodAnchor preferenceGap lower cutoff anchor →
lower < upper →
0 < delta →
delta ≤ 1 / 2 →
√(6 * ↑(Fintype.card Arm) * Real.log ↑(Fintype.card Arm)) < ↑cutoff →
↑cutoff / ↑(Fintype.card Arm) ≤ delta →
let maxBatch : ℕ := FalahatgarEtAl2017MaxingRanking.sourcePruneMaxBatch lower upper delta;
have hbudget :
  ∀ round < FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm),
  FalahatgarEtAl2017MaxingRanking.fixedSampleBudget lower upper
  (FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta delta round) ≤
  maxBatch :=
FalahatgarEtAl2017MaxingRanking.fixedSampleBudget_le_sourcePruneMaxBatch_of_source_round lower upper
delta;
1 - delta / 2 ≤
AppliedModelingLib.pmfProb
(FalahatgarEtAl2017MaxingRanking.freshStoppedPruneTraceStateLaw preferenceGap lower
(FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneContractionFactor (Fintype.card Arm) cutoff delta)
cutoff anchor initial
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneOutcomeLaw preferenceGap hprobability
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) anchor lower upper
delta maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.canonicalFreshPruneDecision
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) lower upper delta
maxBatch hbudget)
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)))
fun
(stateFailure :
  FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceState Arm
  (FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) ×
  Bool) =>
stateFailure.1.1.card ≤ 2 * cutoff ∧
↑(FalahatgarEtAl2017MaxingRanking.stoppedPruneTraceComparisonCount
(FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount (Fintype.card Arm)) cutoff lower upper
delta stateFailure.1.2) ≤
FalahatgarEtAl2017MaxingRanking.sourceLemma5PruneComparisonRateBound (Fintype.card Arm) lower
upper delta

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma15_pruneMultiRound_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: The checked statement retains Lemma 15's strict cutoff, confidence, success probability, and comparison-rate conclusion. Its output endpoint is at most $2n'$ because Algorithm 2 can stop at equality.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 1051-1054

Verbatim source input

Lemma 16. If a , the output of step 6 in $O_{PT-M} \text{AXIMIZE}(S, \epsilon, \delta)$ is a 2ϵ -maximum of S , then w.p. $\geq 1 - 4\epsilon$, $O_{PT-M} \text{AXIMIZE}(S, \epsilon, \delta)$ outputs a .

Expanded PaperInterface specification

```
∀ {Arm : Type u_1} [inst : Fintype Arm] [Nonempty Arm] (anchor fallback : Arm) (candidates : Finset Arm)
  (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon delta : ℝ),
  (∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
  FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap (2 * epsilon / 3) anchor →
  0 < epsilon →
  0 < delta →
  delta ≤ 1 →
  have x : DecidableEq Arm := Classical.decEq Arm;
  1 - delta / 4 ≤
  AppliedModelingLib.PreferenceRL.pmfProbClassical
    (FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw anchor fallback candidates
     preferenceGap hprobability (2 * epsilon / 3) epsilon (delta / 4 / ↑(Fintype.card Arm)))
  fun (output : Arm) => output = anchor
```

Retained governing declarations

- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. parameter: Arm
5. parameter: Arm
6. parameter: Finset Arm
7. parameter: Arm → Arm → ℝ
8. assumption: FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap
9. parameter: ℝ
10. parameter: ℝ
11. assumption: ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second
12. assumption: FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap (2 * epsilon / 3) anchor
13. assumption: 0 < epsilon
14. assumption: 0 < delta
15. assumption: delta ≤ 1
16. conclusion: 1 - delta / 4 ≤
  AppliedModelingLib.PreferenceRL.pmfProbClassical
    (FalahatgarEtAl2017MaxingRanking.canonicalFreshFinalCheckOutputLaw anchor fallback candidates preferenceGap
     hprobability (2 * epsilon / 3) epsilon (delta / 4 / ↑(Fintype.card Arm)))
  fun (output : Arm) => output = anchor
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma16_finalCheckAnchor_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target states the source Lemma 16 branch directly: if the Algorithm-3 anchor is a $2\epsilon/3$ -maximum, every final Compare accepts it jointly with probability at least $1 - \delta/4$, so the canonical final-check output is that anchor.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 1071-1077

Verbatim source input

Lemma 17. For $\delta \geq \frac{1}{\sqrt{n}}$, w.p. $\geq 1 - \delta/2$, steps 1-6 in O PT-M AXIMIZE(S, ϵ, δ) uses $O(\epsilon n^2 \log \frac{1}{\delta})$ comparisons, outputs a set S' of size at most $24n \log n$ and either a is a $2\epsilon/3$ -maximum element or S' contains the absolute maximum element.

$$\frac{1}{\sqrt{n \log n}}$$

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (epsilon delta : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hantisymmetric : ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second),
(∀ (arm : Arm), preferenceGap arm arm = 0) →
  ∀ (hsst : FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap),
  0 < epsilon →
  ∀ (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1),
  1 / ↑(Fintype.card Arm) ≤ delta →
  have maximum : Arm :=
    Classical.choose
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_1 preferenceGap hantisymmetric hsst);
  let cutoff : ℕ := FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff (Fintype.card Arm);
  let maxBatch : ℕ :=
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch (epsilon / 3) (2 * epsilon / 3)
    delta;
  1 - delta / 2 ≤
    FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceSuccessProbability cutoff
    (delta / 4) (delta / 4) (epsilon / 3) (2 * epsilon / 3) maxBatch preferenceGap maximum hprobability
    FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_1
    (FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_2 delta hdelta)
    (FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_3 delta hdeltaLeOne)
    (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_3 epsilon delta) ∧
  (fun (armCount : ℕ) =>
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope armCount (epsilon / 3)
    (2 * epsilon / 3) epsilon delta) = O[Filter.atTop]
  fun (armCount : ℕ) => ↑armCount * (1 + Real.log (1 / delta)) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.adaptivePruneRoundDelta](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceSuccessProbability](#)
- [FalahatgarEtAl2017MaxingRanking.fixedSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope](#)
- [FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff](#)
- [FalahatgarEtAl2017MaxingRanking.sourcePruneRoundCount](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. conclusion: ∀ (epsilon delta : ℝ) (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(hantisymmetric : ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second),
(∀ (arm : Arm), preferenceGap arm arm = 0) →
  ∀ (hsst : FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap),
  0 < epsilon →
  ∀ (hdelta : 0 < delta) (hdeltaLeOne : delta ≤ 1),
  1 / ↑(Fintype.card Arm) ≤ delta →
  have maximum : Arm :=
    Classical.choose
      (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_1 preferenceGap hantisymmetric hsst);
  let cutoff : ℕ := FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeCutoff (Fintype.card Arm);
  let maxBatch : ℕ :=
    FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3PruneMaxBatch (epsilon / 3) (2 * epsilon / 3)
```

```

delta;
1 - delta / 2 ≤
  FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeAnchorPruneTraceSourceSuccessProbability cutoff
  (delta / 4) (delta / 4) (epsilon / 3) (2 * epsilon / 3) maxBatch preferenceGap maximum hprobability
  FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_1
  (FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_2 delta hdelta)
  (FalahatgarEtAl2017MaxingRanking.lemma17_anchorPruneSpec._proof_3 delta hdeltaLeOne)
  (FalahatgarEtAl2017MaxingRanking.theorem6_optMaximizeSpec._proof_3 epsilon delta) ∧
(fun (armCount : ℕ) =>
  FalahatgarEtAl2017MaxingRanking.sourceOptMaximizeAlgorithm3RateEnvelope armCount (epsilon / 3)
  (2 * epsilon / 3) epsilon delta) = O[Filter.atTop]
fun (armCount : ℕ) => ↑armCount * (1 + Real.log (1 / delta)) / epsilon ^ 2

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma17_anchorPrune_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The checked statement is the stated first OPT-Maximize phase: under delta at least 1/n it gives probability 1-delta/2, the square-root retained-size bound, the anchor-or-maximum conclusion, and the displayed asymptotic rate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 1091-1094

Verbatim source input

Lemma 18. For $\delta \geq n_1$, if a , the output of step 6 and S' , the output of step 7 are such that either a is $2\epsilon/3$ -maximum of S or S' contains the absolute maximum element of S , then steps 8-13 of O PT-M AXIMIZE(S , ϵ , δ)

uses $O(|S\epsilon 2| \log n\delta)$ comparisons and w.p. $\geq 1 - \delta/2$, outputs an ϵ -maximum.

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (anchor : Arm) (candidates : Finset Arm)
  (epsilon delta : ℝ) (preferenceGap : Arm → Arm → ℝ)
  (hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap),
  (∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
  (∀ (arm : Arm), preferenceGap arm arm = 0) →
  FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
  0 < epsilon →
  0 < delta →
  delta ≤ 1 →
  1 / ↑(Fintype.card Arm) ≤ delta →
  (FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap (2 * epsilon / 3) anchor v
    ∃ (maximum : Arm),
      FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum ∧ maximum ∈ candidates) →
  have x : DecidableEq Arm := Classical.decEq Arm;
  (1 - delta / 2 ≤
    AppliedModelingLib.PreferenceRL.pmfProbClassical
      (FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTailSourceLaw anchor candidates
        preferenceGap hprobability epsilon delta)
    fun (fallbackOutput : Arm × Arm) =>
      FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap epsilon fallbackOutput.2) ∧
  ↑(FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailComparisonCap candidates (2 * epsilon / 3)
    epsilon (delta / 4 / ↑(Fintype.card Arm)) epsilon (delta / 4)) ≤
  FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailAlgorithm3SourceBudget candidates
  (2 * epsilon / 3) epsilon epsilon delta
```

Retained governing declarations

- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTailSourceLaw](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailAlgorithm3SourceBudget](#)
- [FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailComparisonCap](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. parameter: Arm
5. parameter: Finset Arm
6. parameter: ℝ
7. parameter: ℝ
8. parameter: Arm → Arm → ℝ
9. assumption: FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap
10. assumption: ∀ (first second : Arm), preferenceGap second first = -preferenceGap first second
11. assumption: ∀ (arm : Arm), preferenceGap arm arm = 0
12. assumption: FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap
13. assumption: 0 < epsilon
14. assumption: 0 < delta
15. assumption: delta ≤ 1
16. assumption: 1 / ↑(Fintype.card Arm) ≤ delta
17. assumption: FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap (2 * epsilon / 3) anchor v
  ∃ (maximum : Arm), FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum ∧ maximum ∈ candidates
18. conclusion: (1 - delta / 2 ≤
  AppliedModelingLib.PreferenceRL.pmfProbClassical
    (FalahatgarEtAl2017MaxingRanking.canonicalOptMaximizeTailSourceLaw anchor candidates preferenceGap hprobability
      epsilon delta)
  fun (fallbackOutput : Arm × Arm) =>
    FalahatgarEtAl2017MaxingRanking.EpsilonMaximum preferenceGap epsilon fallbackOutput.2) ∧
  ↑(FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailComparisonCap candidates (2 * epsilon / 3) epsilon
    (delta / 4 / ↑(Fintype.card Arm)) epsilon (delta / 4)) ≤
  FalahatgarEtAl2017MaxingRanking.optMaximizeFinalTailAlgorithm3SourceBudget candidates (2 * epsilon / 3) epsilon
  epsilon delta
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma18_optMaximizeTail_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target formalizes the exact Lemma 18 conditional branch: assuming a $2\epsilon/3$ anchor or a candidate set containing an absolute maximum, steps 8--13 have their subset-dependent finite budget and output an epsilon-maximum with probability at least $1-\delta/2$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 1176-1177

Verbatim source input

Lemma 19. E STIMATE -P ROBABILITY(i, j, ϵ, δ) uses $2\epsilon 12 \log 2\delta$ comparisons and w.p. $\geq 1 - \delta$ approximates $\pi_{i,j}$ to an additive error of ϵ .

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap)
(challenger incumbent : Arm) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
(1 / (2 * epsilon ^ 2) * Real.log (2 / delta) ≤
↑(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) ∧
↑(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) <
1 / (2 * epsilon ^ 2) * Real.log (2 / delta) + 1) ∧
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability challenger incumbent
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta)).toMeasure.real
{batch : Fin (FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) → Bool |
¬|FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta))
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) batch -
(1 / 2 + preferenceGap challenger incumbent)| <
epsilon} ≤
delta
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Arm → Arm → ℝ
3. assumption: FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap
4. parameter: Arm
5. parameter: Arm
6. parameter: ℝ
7. parameter: ℝ
8. assumption: 0 < epsilon
9. assumption: 0 < delta
10. assumption: delta ≤ 1
11. conclusion: (1 / (2 * epsilon ^ 2) * Real.log (2 / delta) ≤
↑(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) ∧
↑(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) <
1 / (2 * epsilon ^ 2) * Real.log (2 / delta) + 1) ∧
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchLaw preferenceGap hprobability challenger incumbent
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta)).toMeasure.real
{batch : Fin (FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) → Bool |
¬|FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalComparisonBatchObservation
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta))
(FalahatgarEtAl2017MaxingRanking.estimateProbabilitySampleBudget epsilon delta) batch -
(1 / 2 + preferenceGap challenger incumbent)| <
epsilon} ≤
delta
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma19_estimateProbability_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target retains Lemma 19's PDF coefficient ceiling($1/(2 \epsilon^2 \log(2/\delta))$), its less-than-one ceiling correction, and the probability-at-least-1-delta additive-epsilon estimate under the paper's Bernoulli comparison law. The coefficient is the documented PDF reading, not a changed claim.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 1202-1203

Verbatim source input

Lemma 20. S TRONG -T RANSITIVITY-R ANKING(S, ϵ , δ) uses $O(n \epsilon^2 \log n \delta)$ comparisons and w.p. $\geq 1 - \delta$ returns an ϵ -ranking.

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon delta : ℝ)
(sourceRanking : Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.PreferenceRanking preferenceGap sourceRanking →
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
0 < Fintype.card Arm →
0 < epsilon →
0 < delta →
delta ≤ 1 →
1 - delta ≤
AppliedModelingLib.PreferenceRL.pmfProbClassical
(FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityBatchLaw preferenceGap hprobability
epsilon delta)
fun
(batchTable :
FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
Fin
(FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm)
epsilon delta) →
Bool) =>
FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking preferenceGap epsilon
(FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput
(FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate epsilon delta
batchTable)
epsilon)) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑(FalahatgarEtAl2017MaxingRanking.strongTransitivityComparisonCount epsilon delta) ≤
5 * ↑armCount ^ 2 * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2
```

Retained governing declarations

- [AppliedModelingLib.PreferenceRL.pmfProbClassical](#)
- [FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking](#)
- [FalahatgarEtAl2017MaxingRanking.PreferenceRanking](#)
- [FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityComparisonCount](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput](#)
- [FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.CenteredComparisonProbabilities preferenceGap) (epsilon delta : ℝ)
(sourceRanking : Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.PreferenceRanking preferenceGap sourceRanking →
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
0 < Fintype.card Arm →
0 < epsilon →
0 < delta →
delta ≤ 1 →
1 - delta ≤
AppliedModelingLib.PreferenceRL.pmfProbClassical
(FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityBatchLaw preferenceGap hprobability
epsilon delta)
```



```

fun
  (batchTable :
    FalahatgarEtAl2017MaxingRanking.StrongTransitivityPairIndex Arm →
    Fin
    (FalahatgarEtAl2017MaxingRanking.strongTransitivitySampleBudget (Fintype.card Arm)
      epsilon delta) →
    Bool) =>
  FalahatgarEtAl2017MaxingRanking.EpsilonPreferenceRanking preferenceGap epsilon
  (FalahatgarEtAl2017MaxingRanking.strongTransitivityRankingOutput
    (FalahatgarEtAl2017MaxingRanking.canonicalStrongTransitivityRankingEstimate epsilon delta
      batchTable)
    epsilon) ∧
  ∀ (armCount : ℕ) (epsilon delta : ℝ),
    0 < armCount →
    0 < epsilon →
    epsilon ≤ 1 →
    0 < delta →
    delta ≤ 1 →
    ↑ (FalahatgarEtAl2017MaxingRanking.strongTransitivityComparisonCount epsilon delta) ≤
    5 * ↑ armCount ^ 2 * (1 + Real.log (↑ armCount / delta)) / epsilon ^ 2

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma20_strongTransitivityRanking_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target preserves Lemma 20's once-per-unordered-pair estimation scheme, epsilon-ranking success probability $1 - \delta$, and $O(n^2 \epsilon^{-2} \log(n/\delta))$ source rate. It uses the proof-consistent complementary reverse estimate and $\epsilon/2$ threshold already documented for the Algorithm 7 display.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 1242-1243

Verbatim source input

Lemma 21. E STIMATE -B ORDA -S CORE(S, a, ϵ , δ) uses $2\epsilon 12 \log 2\delta$ comparisons and w.p. $\geq 1 - \delta$ approximates s(a) to an additive error of ϵ .

Expanded PaperInterface specification

```

∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (arm : Arm)
(epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
(1 / (2 * epsilon ^ 2) * Real.log (2 / delta) ≤
↑(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) ∧
↑(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) <
1 / (2 * epsilon ^ 2) * Real.log (2 / delta) + 1) ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) →
Bool |
¬|FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) arm)
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) labelTable -
FalahatgarEtAl2017MaxingRanking.bordaScore winProbability arm| <
epsilon} ≤
delta
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
- [FalahatgarEtAl2017MaxingRanking.bordaScore](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation](#)
- [FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. conclusion: ∀ (winProbability : Arm → Arm → ℝ) (hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability)
(arm : Arm) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
(1 / (2 * epsilon ^ 2) * Real.log (2 / delta) ≤
↑(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) ∧
↑(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) <
1 / (2 * epsilon ^ 2) * Real.log (2 / delta) + 1) ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) →
Bool |
¬|FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) arm)
(FalahatgarEtAl2017MaxingRanking.estimateBordaScoreSampleBudget epsilon delta) labelTable -
FalahatgarEtAl2017MaxingRanking.bordaScore winProbability arm| <
epsilon} ≤
delta
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.lemma21_estimateBordaScore_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target is Lemma 21's uniform-opponent Borda estimator with the PDF ceiling($1/(2 \epsilon^2) \log(2/\delta)$), additive-epsilon success event, and 1-delta probability. It keeps the source Borda score and finite canonical sampling law explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 1258-1263

Verbatim source input

Theorem 22. B ORDA -R ANKING(S, ϵ, δ) uses $2n$
 $\frac{\epsilon^2}{\log 2n}$
 $\frac{6}{\delta}$
comparisons and w.p. $\geq 1 - \delta$ outputs an ϵ -Borda
ranking.

Expanded PaperInterface specification

```
(∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
have x : DecidableEq Arm := Classical.decEq Arm;
∃ (rankingRule : (Arm → ℝ) → Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule rankingRule ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon
delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) →
Bool |
¬FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking winProbability epsilon
(rankingRule fun (arm : Arm) =>
FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) arm)
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta)
labelTable)} ≤
delta) ∧
∀ (armCount : ℕ) (epsilon delta : ℝ),
0 < armCount →
0 < epsilon →
epsilon ≤ 1 →
0 < delta →
delta ≤ 1 →
↑(FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount armCount epsilon delta) ≤
3 * ↑armCount * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities](#)
- [FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking](#)
- [FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule](#)
- [FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore](#)
- [FalahatgarEtAl2017MaxingRanking.bordaSampleBudget](#)
- [FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw](#)
- [FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Arm : Type u_1} [inst : Fintype Arm] [inst_1 : Nonempty Arm] (winProbability : Arm → Arm → ℝ)
(hprobability : FalahatgarEtAl2017MaxingRanking.BordaWinProbabilities winProbability) (epsilon delta : ℝ),
0 < epsilon →
0 < delta →
delta ≤ 1 →
have x : DecidableEq Arm := Classical.decEq Arm;
∃ (rankingRule : (Arm → ℝ) → Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.IsEmpiricalBordaRankingRule rankingRule ∧
(FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchLaw winProbability hprobability
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon
delta)).toMeasure.real
{labelTable :
FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchCoordinate Arm
(FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) →
Bool |
¬FalahatgarEtAl2017MaxingRanking.EpsilonBordaRanking winProbability epsilon
(rankingRule fun (arm : Arm) =>
```

```

    FalahatgarEtAl2017MaxingRanking.bordaEmpiricalScore
    (FalahatgarEtAl2017MaxingRanking.canonicalBordaBatchObservation
    (FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta) arm)
    (FalahatgarEtAl2017MaxingRanking.bordaSampleBudget (Fintype.card Arm) epsilon delta)
    labelTable)} ≤
    delta) ∧
  ∀ (armCount : ℕ) (epsilon delta : ℝ),
    0 < armCount →
    0 < epsilon →
    epsilon ≤ 1 →
    0 < delta →
    delta ≤ 1 →
    ↑(FalahatgarEtAl2017MaxingRanking.bordaTotalSampleCount armCount epsilon delta) ≤
    3 * ↑armCount * (1 + Real.log (↑armCount / delta)) / epsilon ^ 2

```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.theorem22_bordaRanking_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target preserves Supplement Theorem 22's Algorithm-9 epsilon-Borda-ranking success event at probability $1 - \delta$ and its displayed $2n \epsilon^{-2} \log(2n/\delta)$ count, represented with the finite ceiling-aware source-rate envelope and explicit empirical ranking rule.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 90-91

Verbatim source input

$p_{i,k} \geq \max(p_{i,j}, p_{j,k})$. By simple induction, every SST model has a maximum element and a ranking. And one additional property, that is perhaps more difficult to justify, has proved helpful in

Expanded PaperInterface specification

```
∀ {Arm : Type u_1} [inst : Fintype Arm] [Nonempty Arm],
have x : DecidableEq Arm := Classical.decEq Arm;
∀ (preferenceGap : Arm → Arm → ℝ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
(∃ (maximum : Arm), FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum) ∧
(∃ (ranking : Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.PreferenceRanking preferenceGap ranking)
```

Retained governing declarations

- [FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum](#)
- [FalahatgarEtAl2017MaxingRanking.PreferenceRanking](#)
- [FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Fintype Arm
3. assumption: Nonempty Arm
4. conclusion: ∀ (preferenceGap : Arm → Arm → ℝ),
(∀ (first second : Arm), preferenceGap second first = -preferenceGap first second) →
FalahatgarEtAl2017MaxingRanking.StrongStochasticTransitivity preferenceGap →
(∃ (maximum : Arm), FalahatgarEtAl2017MaxingRanking.AbsoluteMaximum preferenceGap maximum) ∧
(∃ (ranking : Fin (Fintype.card Arm) → Arm),
FalahatgarEtAl2017MaxingRanking.PreferenceRanking preferenceGap ranking)
```

Lean proof endpoint (built separately)

FalahatgarEtAl2017MaxingRanking.sstMaximumAndRankingExistence_proof

Recorded source-to-Spec screening Verdict: matches

Reason: The target directly expresses the source's simple finite-SST induction claim: an antisymmetric finite SST preference model has an absolute maximum and a complete preference ranking. It introduces no additional behavioral or probabilistic premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation